

Polity Size and Local Government Performance: Evidence from India[†]

By VEDA NARASIMHAN AND JEFFREY WEAVER*

Developing countries have increasingly decentralized power to local governments. This paper studies the implications of a central element of decentralization (polity size) using population-based discontinuities that determine local government boundaries for over 100,000 Indian villages. Over the short and long run, individuals allocated into local governments with smaller populations have better access to public goods. We provide suggestive evidence that these results are related to heightened civic engagement and stronger political incentives, but not to other mechanisms such as elite capture. (JEL D72, H41, H75, H76, O17, O18, R50)

Developing countries have increasingly moved toward political decentralization and devolution of power to elected local governments. One key aspect of decentralization is the division of the population into local government units, where the size of these polities may be an important determinant of their performance. A number of models suggest benefits from dividing a population into a larger number of smaller local governments. This may better cater to heterogeneous tastes (Oates 1972) or incentivize public good provision through competition and yardstick comparisons (Tiebout 1956; Besley and Case 1995). Smaller jurisdictions are likely to be more homogeneous, which may reduce conflict (Mansuri and Rao 2013; Bazzi and Gudgeon 2021) and may make it easier to monitor leaders and hold them accountable (Seabright 1996; Boffa, Piolatto, and Ponzetto 2016).

Alternatively, there are many potential drawbacks to smaller polities. Smaller jurisdictions may be more easily captured by elites (Ambedkar 1932; Bardhan and Mookherjee 2000; Bardhan 2002), miss out on economies of scale (Alesina and Spolaore 1997), or fail to internalize externalities (Lipscomb and Mobarak 2016).

*Narasimhan: Department of Economics, New York University-Abu Dhabi (email: veda.narasimhan@nyu.edu). Weaver: Department of Economics, University of Southern California (email: jbweaver@usc.edu). Rema Hanna was the coeditor for this article. We are grateful for helpful comments from the editor and anonymous referees, Vittorio Bassi, David Blakeslee, Bruno Caprettini, Lorenzo Casaburi, Ernesto Dal Bo, Ray Fisman, Macartan Humphreys, Matt Kahn, Korhan Kocak, Guilherme Lichand, Samreen Malik, Paul Niehaus, Paulina Oliva, Ajay Shenoy, Joachim Voth, and David Yanagizawa-Drott, as well as seminar audiences at American University, Brown, Dartmouth, ETH Zurich, Georgetown, National University of Singapore, NBER Summer Institute (Development), NYU-Abu Dhabi, PacDev, Stockholm University, SoCCAM, University of British Columbia, University of California-San Diego, University of Manchester, University of Southern California, University of St. Gallen, University of Washington, University of Zurich, and Yale University. All views expressed are those of the authors. Funding for this project was provided by the Swiss National Science Foundation (grant 187942) and the Forschungskredit of the University of Zurich (grant FK-19-017). This paper subsumes and replaces our previous work in Narasimhan (2021) and Weaver (2021).

[†]Go to <https://doi.org/10.1257/aer.20221712> to visit the article page for additional materials and author disclosure statement(s).

A larger unit may also have a larger pool of talent for political leadership and could have more political competition, which puts greater pressure on leaders to perform (Gerring et al. 2015). Given the many plausible theoretical mechanisms pushing in different directions, it is unclear how polity size will affect government performance. Understanding these consequences is essential for creation of political jurisdictions as well as relevant in evaluating the empirical relevance of different theories of decentralization centered around polity size.

This paper provides causal evidence on the consequences of polity size in the context of the Indian state of Uttar Pradesh. Uttar Pradesh has a population of 241 million people and would be the fifth-most populous country in the world if it were an independent country. As in the rest of India, the village council, or gram panchayat (GP), is the lowest level of elected government. Each GP typically contains multiple villages, with local representatives elected from each village and one head (*pradhan*) elected for the entire GP. The GP oversees the provision of local public goods such as education and sanitation, is a primary implementer of a large workfare program (National Rural Employment Guarantee Scheme, abbreviated as NREGS), and plays a role in many welfare programs. Outside of those formal responsibilities, GP leaders play a crucial role in helping their constituents access benefit programs and lobbying bureaucrats and politicians for the allocation of resources to their GPs.

Our empirical approach takes advantage of a unique rule in Uttar Pradesh that determines the allocation of villages to local governments. This rule stipulates that based on the most recent decadal census, villages whose population exceeds 1,000 should be allocated into their own GP. Uttar Pradesh has conducted two rounds of panchayat delimitation based on this rule—once in 1995 and a second time in 2015—and boundaries have otherwise remained mostly fixed, even if a village's population exceeded the threshold at some time between 1995 and 2015. The rule causes villages with populations just below 1,000 to end up in much more populous GPs than villages with populations just above the cutoff; in the 2015 delimitation, villages just below the cutoff ended up in GPs with an average population that is roughly 40 percent larger than villages above the cutoff. We use this in a regression discontinuity (RD) design to estimate the effect of allocation into a smaller GP.

We combine census and administrative datasets spanning a 30-year period to evaluate the effect of this allocation on public service delivery, including novel village-level data on public benefit receipt. We first investigate village-level infrastructure and find that villages allocated to smaller GPs have significantly better access to educational facilities in both the short and long run, which results in economically meaningful gains in educational attainment. There are also increases in all-weather roads and fair price shops (which provide access to subsidized food), but we do not see effects on piped water.

Next, we examine individually targeted benefit programs. Households in villages allocated to smaller polities are more likely to have toilets, closed drainage systems, and houses made out of high-quality materials like brick rather than organic materials. These effects are sizable; for example, there is an 18 percent increase in the likelihood of having a toilet with a corresponding decrease in open defecation. This effect is likely due to the improved implementation of programs that fund housing and sanitation improvements, for which local leaders play a substantial role. We also find elevated access to eight benefit programs ranging from

pensions to government-provided health insurance, which is consistent with leaders in smaller polities putting more effort into helping citizens make claims on the state. Furthermore, we find that benefits are also better targeted in smaller polities, with a stronger correlation between poverty and possession of Below Poverty Line (BPL) ration cards among villages to the right of the cutoff.

Finally, we use administrative data to measure the implementation of the NREGS workfare program. Individuals in villages allocated to smaller GPs have better workfare access, including NREGS earnings that are around 16 percent higher. These data also disaggregate workfare participation by caste and gender, and we find that effects are similar across advantaged and disadvantaged groups.

The broad array of outcomes that we study helps rule out concerns about multitasking (Holmström and Milgrom 1991), such as the improvements in infrastructure coming at the expense of welfare or workfare programs. We also conduct a large number of robustness checks to show that other government programs do not use the same population cutoff and that our estimates are not the result of mechanical funding rules favoring smaller polities. We additionally do not detect negative spillovers onto untreated villages.

These estimates pertain to a local average treatment effect of shifting citizens into GPs with approximately 1,000 residents from a range of counterfactual populations. However, policymakers may be interested in the treatment effect of specific policies (e.g., shifting citizens from a GP of 6,000 residents to one of 1,000 as compared to a shift from 3,000 to 1,000) or in polity sizes other than 1,000. We leverage the same delimitation rule to estimate these additional treatment effects. For the intuition of this exercise, suppose that villages *a* and *b* were in a GP with village *c*, but village *c*'s population is around the population cutoff at the time of delimitation. Under the delimitation rules, if village *c* is above the cutoff, it will be split off, and villages *a* and *b* will remain together in a smaller GP; if below, then villages *a*, *b*, and *c* will remain together in a larger GP.

We first use this to disaggregate the previously estimated treatment effects. The GP population for *c* will be approximately 1,000 if it were split off. If it is not, its GP population will be the sum of the populations of villages *a*, *b*, and *c*, which we denote as *x*. We reestimate the RD for different values of *x* to observe the consequences of shifting to a GP population of 1,000 from different counterfactual populations *x*. We find evidence that the treatment effect is greater when *x* is larger. Second, we generate an alternative RD design, with village *c*'s population as a running variable for the size of the polity to which villages *a* and *b* are allocated. This estimates the treatment effect of going from a GP of population *x* to one of $x - 1,000$. The effects appear to be stronger in cases when the created GPs have populations of 1,000 to 3,000. This is consistent with the effect coming from creating polities of certain sizes rather than generically making them smaller.

To better understand the drivers of the effects we find, we test a set of potential mechanisms from the theoretical decentralization literature. These tests provide suggestive evidence on why there is an increase in service delivery outcomes, as well as being of intrinsic interest in providing evidence on mechanisms that have previously been difficult to document empirically. The strongest support appears to be for political channels. We show that important forms of civic engagement are greater in smaller polities and that candidates and elected representatives are somewhat less

likely to have a criminal record. Additionally, we find no evidence of greater levels of elite capture (Bardhan and Mookherjee 2000), where high-caste and wealthy individuals are no more likely to be elected and the increases in service access are similar across high- and low-caste groups. Finally, there is evidence that effects are stronger in villages that did not have the largest population in their pre-delimitation GP, and so delimitation generates a larger increase in their political power.¹

These results have implications for decentralization policy and creation of local governments. Most directly, these findings measure the consequences of reductions in local government size in India, where average GP population varies massively across states. They also provide principles for the creation of smaller local government bodies across a broader global context. Policies reforming polity size are especially exciting since they are feasible to implement at scale even in the context of low-capacity states.

The paper contributes to three main strands of the literature. First, we provide novel estimates of the causal effect of polity size on government performance. Much of the work on optimal government scale has been theoretical due to the difficulty of finding plausibly exogenous sources of variation in the size of political units (Alesina and Spolaore 1997; Boffa, Piolatto, and Ponzetto 2016; Faguet 2004). The most closely related empirical work uses difference-in-differences to study the consequences of splits or amalgamation of municipalities or districts (e.g., Lassen and Serritzlew 2011; Pierskalla 2016; Dahis and Szerman 2021; Singhania 2022; Cassidy and Velayudhan 2022; Cohen 2023). Our approach expands upon that approach by measuring not only the average effects of amalgamation or fragmentation but the consequences of specific polity sizes. This provides more implementable guidance for local government creation, especially since one might expect nonlinearities in the polity size-performance relationship such that average effects are not necessarily informative. For example, it is unlikely that more splits are always better; jurisdictions that each contain only a single person are almost certainly suboptimal.² These findings provide lessons that can potentially generalize to local government design in settings outside of north India.

Second, we complement the theoretical literature on political decentralization by examining mechanisms through which polity size may affect outcomes. Polity size is a key element of many of the most prominent theories of decentralization, but empirically testing these theories is challenging. We find some suggestive evidence in favor of increased political participation and better-aligned political incentives in smaller local governments. This is consistent with Seabright (1996) and Cremer, Estache, and Seabright (1994) among others, who argue that decentralization helps to ease some of the issues associated with political accountability in democracies. We do not find strong support for many other plausible mechanisms in our context,

¹ We also check whether the observed effects can be explained by creating more homogeneous or geographically compact GPs. We find no evidence for either channel.

² The RD approach likely also provides more generalizable estimates by studying changes induced by an exogenous cutoff. Difference-in-differences estimates are typically based around politically motivated splits/amalgamations and so naturally may be concentrated among a subset of changes in which splits generate benefits (Pierskalla 2016). Previous papers have also focused on much larger levels of aggregation, such as districts in Indonesia.

such as elite capture or lower identity group fractionalization (Alesina, Baqir, and Easterly 1999; Bazzi and Gudgeon 2021).

Third, we contribute to the literature linking design of local government to public goods provision in developing countries. A growing literature has explored aspects such as democratization (Olken 2010; Martinez-Bravo et al. 2022), representative identity (Chattopadhyay and Duflo 2004; Martinez-Bravo 2014), and technological solutions to governance problems (Lewis-Faupel et al. 2016; Banerjee et al. 2020). We complement these papers by studying a different aspect of the design of local government. A nice feature of polity size reforms is that they require minimal state capacity to implement, as evidenced by these effects in one of the lowest-capacity states in India.

The remainder of the paper is organized as follows. Section I discusses the institutional background, while Section II describes the data. Section III explains the empirical approach, Section IV provides results, and Section V investigates mechanisms. Section VI concludes and discusses the policy implications.

I. Background

A. History of Local Governance in India

Rural village councils, known as gram panchayats, have existed in some form in India since the premodern era. The first state-level legislation concerning their administration was introduced soon after independence, but it was not until the 1990s that a consistent national-level policy was established. Until that point, GPs in most states were given limited responsibilities, and government administration was primarily at the level of the block (consisting of 50 to 200 villages) or higher levels of government.

In the 1990s, India and many other large developing countries like China and Indonesia engaged in large-scale devolution of power to elected local governments. In India, this was based on the Seventy-Third Amendment to the Indian Constitution, a comprehensive decentralization reform with three key components: (i) formation of a three-tier system of governance at the village (gram panchayat), subdistrict, and district levels; (ii) devolution of power and responsibility to the gram panchayat; and (iii) standardization of elections at the village level. By formally amending the constitution, the federal government ensured uniformity in these key aspects but allowed states discretion over which responsibilities to devolve to local governments and how to form councils (Chaudhary and Iyer 2022).

Leadership of the local government body consists of ward councilors and a head (*pradhan*). “Wards” typically consist of a few hundred voters, and each elect a single local representative (“ward councilor”). The system for electing *pradhans* varies from state to state; in the state we focus on, there is direct election by citizens at the same time as the ward councilor elections. For both types of positions, candidates are not officially aligned with political parties on the ballot, the candidate with the most votes wins regardless of whether they have a majority, and elections occur every five years with no formal term limits. However, in each election cycle, a third of positions are reserved for women, and as many as half may be reserved for individuals from traditionally disadvantaged identity groups, which imposes *de facto*

term limits that could affect performance (Dal Bó and Rossi 2011; Genicot, Brown, and Kochhar 2021). These leaders are supported by a panchayat secretary, who is the bureaucratic counterpart to the pradhan and carries out village-level bureaucratic tasks. Panchayat secretaries in the state we study are typically assigned to multiple GPs, with each secretary serving an average of more than 10 GPs and 15,000 citizens.

The Seventy-Third amendment lists 29 areas that can be devolved as responsibilities of local governments (see online Appendix Table A1 for a full list) but lets states decide which to devolve (Chaudhary and Iyer 2022). These include important areas such as education, sanitation, and drinking water. The GP also administers local development programs targeted at rural areas, where the most important of these is the National Rural Employment Guarantee Scheme, a large workfare program.

Beyond oversight of those areas, a large literature in political science further documents the role of GP leaders in helping citizens secure individual-level benefits. First, GP leaders often provide information to citizens on the existence of government programs, practical knowledge on how to access them, and assistance with preparing applications. Second, GP leaders frequently lobby and make calls to bureaucrats or other important actors to exert informal influence over access to these benefits.³ Third, they play a role in program targeting and verification for numerous programs, where their signature or stamp is required as part of some applications for identity verification (Chauchard 2017; Bussell 2019). Through these roles, they can substantively impact the level of services available to their constituents.

Different states have taken varied approaches in creating GPs. Average GP population across India is just under 3,500 people, but this varies substantially. At one extreme, gram panchayats in the states of Bihar, Kerala, and West Bengal are composed of many villages and have average populations of over 10,000 (online Appendix Figure A2). GPs in other states contain an average of less than 2,000 people. These choices are largely a function of historical path dependence (for example, the key rule governing GP size in Uttar Pradesh originates in a law from 1947) and so may not reflect currently optimal policy.

B. Local Governance in Uttar Pradesh

This paper will focus on local governments in Uttar Pradesh, the most populous state in India. Uttar Pradesh is the third-poorest state in India, with 38 percent of its population deemed poor, and has a PPP-adjusted GDP per capita comparable to that of Mali (NITI Aayog 2021). Nearly four-fifths of its population lives in rural areas (Census of India 2011), where public service delivery is among the worst in the country.

Uttar Pradesh was one of the first states in India to introduce panchayats under the Uttar Pradesh Panchayat Raj Act of 1947. However, until the Seventy-Third Amendment in 1993, the panchayat system was widely regarded as nonfunctional (Alsop, Krishna, and Sjoblom 2000). At that point, Uttar Pradesh amended existing laws to fall in line with the constitutional amendment and legally devolved 25 of

³See Chauchard (2017, chap. 3); Bussell (2019, chap. 6); Kruks-Wisner (2018, chap. 4); and Dunning and Nilekani (2013) for multiple examples of GP leaders executing both the first and second roles.

29 possible functionalities to GPs, including education, drinking water, and village development (Uttar Pradesh Panchayat Laws Amendment Act of 1994). However, in practice, the vast majority of these were not actually devolved.⁴ Panel A of online Appendix Table A1 categorizes each of the 29 functionalities into those that were actually devolved, those that were legally but not actually devolved, and those that were not devolved. This points to which outcomes may be affected by reforms to GP polity size and informs our choice of outcomes.

We focus on Uttar Pradesh since it uses explicit population thresholds to allocate villages to local government units. GP boundaries are determined by officials of the State Election Commission, who join proximate villages to create local government units (Jha 2021). A typical GP includes 1 to 4 villages, each of which are themselves a collection of geographically proximate smaller clusters of households administratively called “hamlets.” Section 11-F of the 1994 Uttar Pradesh Panchayat Laws Amendment states that the government will declare “a village or group of villages, having, so far as practicable, a population of one thousand, to be a Panchayat area.” It further defines villages as those recorded in the census of India, states that these census-defined villages should not be divided, and specifies use of population counts from the most recent census. An official following this rule will be more likely to allocate villages with a population of 1,000 or more into a GP of their own, whereas villages below the cutoff will be put in a GP with other villages. Section III will show that officials follow this rule, which we will use to examine the effect of GP size.

Uttar Pradesh has undergone two major rounds of panchayat delimitation. The first occurred in 1995 and was based on data from the 1991 census, leading to the creation of 58,620 gram panchayats. There was a consolidation in 2000 to 52,929 GPs, and the boundaries remained steady for the next 15 years, with 52,001 GPs in the 2005 elections and 51,914 GPs in 2010. In 2015, there was a second major round of delimitation based on data from the 2011 census, and the number of GPs expanded to 59,074 (Uttar Pradesh State Election Commission 2021).⁵ For the remainder of the paper, we will analyze the 1995–2014 and post-2015 periods as distinct eras (see online Appendix Figure A1 for an example of this process).

C. Financing and Allocation of Public Programs

This paper measures how polity size affects public service delivery. To understand these estimates, one must consider the underlying funding and allocation processes. These processes differ across programs (for example, they are different for workfare programs than for school construction) but conceptually can be divided into three categories.

The first category is community-level services for which funding comes from the state or central government but a block- or district-level bureaucrat may determine

⁴ See Ministry of the Panchayati Raj (2016, Tables 3.4A–C) for a list of those that were legally devolved and Indian Institute of Public Administration (2020, Tables 11 and 12) for a list of those that were actually devolved.

⁵ GP boundaries should also have been redrawn at the time of the 2005 elections using 2001 census data. After reading through releases of the State Election Commission and news sources from this time period, we have not found historical documentation of why this did not happen; it could have simply been an oversight by state election officials.

allocations. A typical example would be construction of a primary school, which is funded by the state or central government (e.g., the District Primary Education Program studied in Khanna 2023), but lower-level bureaucrats would manage the process of where schools are built. These decisions are well-known to be influenced by political considerations (Wilkinson 2006; Vaishnav and Sircar 2012), and so the presence of these types of public goods in a community can reflect the effort of local government leaders in lobbying bureaucrats and higher-level politicians.

The second category consists of individual-level benefits and programs that may be initiated or assisted by local governments but are funded by higher levels of government. This includes programs we will study that fund the construction of improved housing and toilets, for which eligible households submit applications to bureaucrats, and the NREGS workfare program, where GP leaders propose projects, hire workers, and generally manage implementation. In these cases, decisions must be approved by bureaucratic actors (see Banerjee et al. 2020 for a description of processes for NREGS), but the effort of GP leaders plays a major role in what gets implemented.

The final category is services initiated by the local government and funded by its budget. Although gram panchayats have the power to levy taxes on certain economic activities, formal tax collection is minimal at the local level as in many other developing countries; across India, local taxes are 0.4 percent of total tax revenues (Rao 2007). Their budget predominantly comes from transfers from the state and central governments, which both are determined by a funding formula from the State Finance Commission (Sethi et al. 2004). Over most of the study period, the formula was linear in GP population (weight of 80 percent) and the area of the GP (20 percent weight); in 2011, this switched to 80 percent based on population and 20 percent based on scheduled caste/scheduled tribe population share (Chakraborty, Gupta, and Thakur 2018). Under these formulas, GPs receive similar per capita transfer amounts; we will later show that these formulas are followed. Over the period 2013–2020, local government budgets in Uttar Pradesh averaged Rs544 (\$8) per inhabitant annually.⁶ Centre for Policy Research (2019) analyses GP budgets across many states and finds that the largest expenditure categories are the creation of “community assets” (28 percent), water and sanitation (22.5 percent), and roads and bridges (20.4 percent).⁷

Local governments can affect each category in different ways. For the first two categories, local leaders can lobby higher-level bureaucrats and politicians for allocations to their GP or to assist constituents seeking individual-level benefits. For the latter two, leader efforts in implementation will affect provision. For all three categories, corruption may be relevant in how spending translates into resources on the ground. Both effort and corruption are plausibly affected by polity size; smaller local governments may be easier for elites to capture but also could have more citizen monitoring that results in less corruption and greater effort. Fundamentally, the net effect of local government size is an empirical question, as it is not obvious which of these and other mechanisms will end up being empirically relevant.

⁶This is similar to the per capita expenditure on NREGS over this period (approximately Rs440 per person), for which funding comes from the Ministry of Rural Development rather than the GP budget.

⁷See panel B of online Appendix Table A1 for a full breakdown of all expenditure categories.

II. Data

This paper relies on data from the Indian census and administrative datasets from different government departments. These data are primarily at the village level, which is advantageous because villages are nested within GPs, and so outcomes are measured consistently regardless of GP assignment. This section gives an overview of each dataset used in the paper, but additional details can be found in the online Appendix.

Census.—We use data from the Census of India (Government of India 1991, 2001, and 2011), and version 1.5 of the Socioeconomic High-resolution Rural-Urban Geographic Dataset (SHRUG) to link villages across the waves of the census (Asher et al. 2021).⁸ We use population information from the 1991 and 2011 Primary Census Abstract to construct the regression discontinuity running variables. Data from the Village Amenities component of the 2001 and 2011 census is used to measure the presence of village-level infrastructure (e.g., all-weather roads, schools, and electricity). We also use the Houselisting tables of the 2011 census to measure the percent of households in the village who have a toilet, have houses built of particular materials, or have access to amenities such as electricity or in-house water supply. In Section IV, we will examine the effect of polity size on the set of outcomes available in the census data that have been devolved to the GP leadership in practice.⁹ We will also report results for services that have been legally but not actually devolved to the GP in the online Appendix for robustness.

Matching Villages to Gram Panchayats.—We match villages to their corresponding GPs based on two sources of data. To match villages to their GP from 2015 to the present, we use the 2021 version of the Local Government Directory from the Ministry of Panchayati Raj (2020). This records the association of each village with its GP as well as the unique village ID code from the 2011 census of India, so it is straightforward to link to census data, with a match rate of 99.13 percent.

To match villages to their GP for the period 1995–2014, we use data from the Department of Drinking Water and Sanitation, which maintains a yearly panel of the association between villages and GPs between 2009 and 2020 (Department of Drinking Water and Sanitation 2011). We matched it to the census data using a multiround matching process combining population data and fuzzy name matching, which gives a match rate of over 96 percent (see online Appendix H for details).

⁸In almost all cases, the unit of a census village remains contiguous over the full period of our panel. Villages are split in less than 1 percent of cases, where these splits are unrelated to the treatment we study. While there is a modest consolidation of villages between 1991 and 2001, this almost entirely affects small villages that are far below the discontinuity we study (89.6 percent of consolidated villages have a population less than 800, and 82.0 percent have a population below 500) so does not affect our results.

⁹We will additionally restrict to looking at outcomes that are found in more than 5 percent and less than 95 percent of untreated villages near the threshold. Some outcomes measured in the census are very uncommon; for example, medical colleges are found in less than 0.06 percent of villages. If a particular type of infrastructure is uncommon, then we are likely to be underpowered to detect plausible effect sizes or will suffer from type M error in cases of a statistically significant estimate (Gelman and Carlin 2014).

Socioeconomic and Caste Census (SECC).—We use village-level data from the SECC to measure educational attainment and poverty (Ministry of Rural Development 2011–2012). The SECC surveyed every household in the country in 2012 and records the number of people in each village by level of educational attainment (e.g., no education, some primary, etc.) as well as by income level. We also use data from Asher et al. (2021), who use household-level SECC data to estimate the fraction of the population living below the poverty line for every village in India.

Mission Antyodaya.—For village-level information on the delivery of public programs in the 2019–2020 period, we use data from the Mission Antyodaya initiative (Ministry of Rural Development 2019). These data were collected as part of a central government initiative and are based primarily on administrative data from government departments; for example, the Department of Food and Civil Supplies reports the number of households in each village who hold BPL ration cards. We focus on the delivery of eight programs measured in these data, which are discussed in more detail in Section IV.

National Rural Employment Guarantee Scheme.—We downloaded publicly available administrative data on the National Rural Employment Guarantee Scheme from the MGNREGA Public Data Portal in March 2021 (Ministry of Rural Development 2016–2020). The Public Data Portal aggregates data on workfare program delivery at an annual level between 2016 and 2020. This includes information such as the number of projects completed, demand for work, days of work provided, wages, and expenditure on materials for each gram panchayat.

Financial Data.—To understand the financial resources available to the panchayats, we use data on GP budgets and spending from the Panchayati Raj accounting system (eGramSwaraj). This contains annual data on spending, transfers from central/state government, and local tax revenues from 2013 through the present for every GP in Uttar Pradesh (Ministry of the Panchayat Raj 2012–2020).

Election and Political Participation Data.—We collected data from the 2010, 2015, and 2020 gram panchayat elections from the Uttar Pradesh State Election Commission (2010, 2015, 2020). For 2015 and 2020, we observe the vote share and characteristics of each contesting candidate, including education, age, gender, and caste background, while for 2010, we observe those for the winner only. In the 2010 elections data, we also observe the home village of the winning candidate. In the 2020 elections, the data contain information on the value of assets held by each candidate as well as any criminal history.¹⁰ We also collected data on participation in public meetings of the local government in 2022–2023 (gram sabhas) from the Ministry of the Panchayati Raj and Ministry of Rural Development (2022–2023).

¹⁰ Wealth is divided into immovable (illiquid assets, such as land or property) and movable wealth (liquid assets, such as vehicles and jewelry). Since there is substantial dispersion and many values of zero, we take the inverse hyperbolic sine when using it. Criminal history is recorded as yes, no, and missing. From examination of the original affidavits, criminal history appears to often be strategically omitted relative to fields like education/wealth. We code no criminal record as a 0, missing as a 1, and confirmed criminal records as a 2 to account for this. See the online Appendix for more details.

These meetings were for the purpose of creating a development plan for the GP, and these data contain information on the number of attendees, broken down by gender and caste category. We also collected personnel records on the allocation of panchayat secretaries to GPs in 2022–2023 from the same ministry.

Combining Data Sources.—The majority of these datasets are at the village level and contain the unique village ID code from the 2011 census so that we can match perfectly across datasets. In cases when that is not available, we use fuzzy name matching combined with data on population counts that significantly improve the matches (see online Appendix H for details). Online Appendix Table A2 shows the match rates between the different datasets and the census data that are the backbone of our approach. Match rates are typically above 95 percent, and, most importantly, likelihood of matching is unrelated to the discontinuity that we study for all of the data (columns 3 and 4) so matching issues will not bias our estimates.

III. Empirical Strategy

We begin with a sharp RD design based around population cutoffs, where villages above the cutoffs are more likely to be assigned to a less populous GP. For GP composition between 1995 and 2014, the cutoff is a population of 1,000 in the 1991 census, while for GP composition after 2015, the cutoff is a population of 1,000 in the 2011 census. The following regression discontinuity specification estimates the impact of delimitation into a smaller GP:

$$(1) Y_{v,g} = \gamma_0 + \gamma_1 \mathbf{1}\{p_{v,g} \geq c\} + \gamma_2 f_1(p_{v,g}) + \gamma_3 f_2(p_{v,g}) \mathbf{1}\{p_{v,g} \geq c\} + \epsilon_{v,g}.$$

$Y_{v,g}$ is the outcome of interest in village v of GP g . $p_{v,g}$ is the population of village v in the relevant pre-delimitation census, while c is the population cutoff used for determination of treatment. The coefficient γ_1 captures the treatment effect, while the functions f_1 and f_2 reflect a continuous but potentially nonparametric relationship between the running variable and the outcome. Using the data-driven approach of Calonico, Cattaneo, and Titiunik (2014) and *rdrobust* command of Calonico et al. (2017), we apply a triangular weighting kernel in distance from the threshold, calculate an MSE-optimal bandwidth with a linear polynomial estimated within the bandwidth on either side of the cutoff, and calculate heterogeneity-robust standard errors clustered at the GP level g .¹¹

Figure 1 tests whether the government follows the delimitation rules, showing the relationship between village population in the census and either (i) the population of that village's GP post-delimitation (panels A and B) or (ii) the likelihood that the village is the only one in its GP (panels C and D). Panels A and C show the relationship for the 1991 village population, while panels B and D plot this for the 2011 village population. Column 1 of Table 1 provides the corresponding estimates in table form: crossing the treatment threshold leads to an average reduction in GP

¹¹ The MSE-optimal bandwidth and polynomial will vary for each outcome Y , following Calonico, Cattaneo, and Farrell (2018). We show robustness to alternative kernels, functional forms, and bandwidths in online Appendix J.

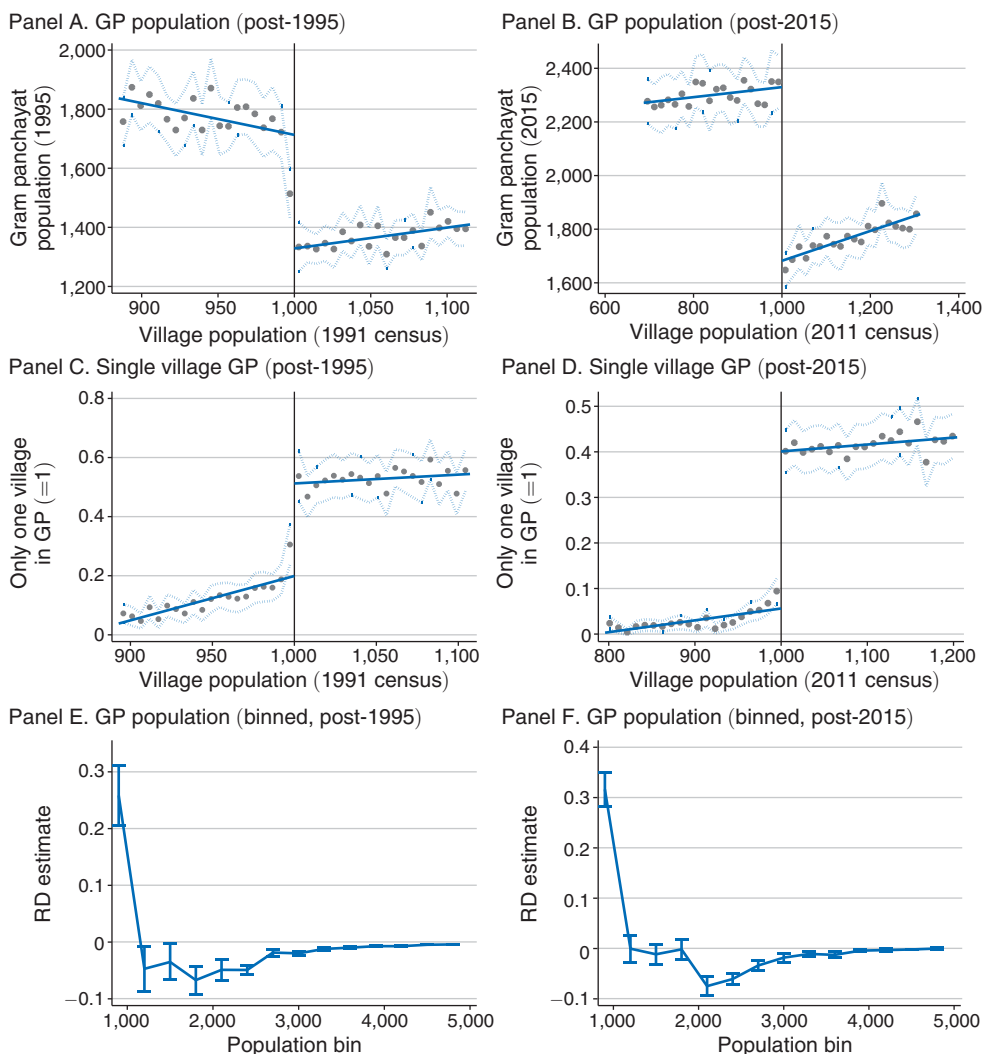


FIGURE 1. FIRST STAGE OF VILLAGE POPULATION ON GRAM PANCHAYAT POPULATION IN UTTAR PRADESH

Notes: These figures plot the first-stage relationship between GP population and village population. Panels A, C, and E plot this for the 1995–2014 GP and village population as measured in the 1991 census, while panels B, D, and F do this for the post-2015 GP and village population from the 2011 census. Panels A and B show the relationship between total GP population and village population. Panels C and D show the relationship between whether a village is the only one in its GP and the village population. Panels E and F rerun the main RD, but where the dependent variable is a dummy variable for whether the GP population that the village is assigned to is in the range (900–1,199), (1,200–1,499), ..., (4,800–5,099). We plot the coefficient and confidence intervals for each of these bins.

size of 362 ($t = 13.1$) persons for the 1991 discontinuity, while the reduction is approximately 649 persons for the 2011 discontinuity ($t = 21.8$). The larger second value reflects the larger average GP size in the later period.¹²

¹²Online Appendix I shows why not all villages with a population over 1,000 are split into separate GPs. In most cases, it is because the village's neighboring villages lack the population to form an independent, contiguous GP themselves.

TABLE 1—FIRST STAGE AND BASELINE OF BASELINE VILLAGE CHARACTERISTICS

	First stage	Baseline covariates (1991)					
	GP pop.	Primary school	Middle school	Paved road	Electrified	SC pop.	Literate pop.
<i>Panel A. First stage and balance of characteristics, 1991</i>							
RD_Estimate	−361.929 (27.643) [0.000]	−0.007 (0.014) [0.600]	0.009 (0.007) [0.215]	0.019 (0.015) [0.189]	0.008 (0.011) [0.435]	−0.007 (0.005) [0.148]	0.001 (0.004) [0.736]
Dep var mean	1,775.423	0.698	0.063	0.399	0.178	0.251	0.274
Bandwidth	115	231	373	283	352	302	185
Effective observations	8,899	18,735	31,050	23,060	29,114	24,743	14,909
	First stage	Baseline covariates (2011)					
	GP pop.	Primary school	Middle school	Paved road	Electrified	SC pop.	Literate pop.
<i>Panel B. First stage and balance of characteristics, 2011</i>							
RD_Estimate	−649.222 (29.830) [0.000]	0.003 (0.012) [0.837]	−0.002 (0.013) [0.862]	0.007 (0.014) [0.588]	0.022 (0.015) [0.129]	0.003 (0.006) [0.630]	0.004 (0.003) [0.262]
Dep var mean	2,298.491	0.750	0.283	0.644	0.572	0.254	0.565
Bandwidth	313	257	337	289	280	270	209
Effective observations	25,059	20,795	26,994	23,173	22,264	21,671	16,859

Notes: This table reports the first-stage estimates and balance checks for both delimitation episodes. Column 1 reports the first-stage estimates of the effect of a village having a pre-delimitation census population above the treatment population threshold (1,000) on the population of that village's associated post-delimitation GP. The remaining columns check whether any village characteristics in the pre-delimitation census change discontinuously at that population threshold. The first four baseline covariates are whether the village has a primary school, middle school, paved road, or electricity, while the remaining two are the fraction of the village's population that is scheduled caste or literate. Panel A runs these analyses for the 1995 delimitation and 1991 census, while panel B does this for the 2015 delimitation and 2011 census. Each specification uses a linear polynomial, triangular kernel, and MSE-optimal bandwidth estimated following Calonico et al. (2017). Standard errors are clustered at the gram panchayat level and reported below the point estimates. *p*-values are reported within brackets.

Crossing the population cutoff of 1,000 leads to changes along multiple different dimensions. The village will not only belong to a GP with a smaller population but also one that is more geographically concentrated and may have a more homogeneous population, among other characteristics. That complicates the interpretation since there are multiple bundled treatments. However, it is still the treatment effect of interest from a policy perspective; any reduction in the size of local political bodies will necessarily entail simultaneous changes along the other dimensions. We thus focus on estimating the sharp discontinuity rather than instrumenting for any one of these treatments in a fuzzy RD design. Section V will return to this point by examining which of these treatments explains the observed effects.

There are two key identifying assumptions for this approach. The first is that the running variable is not manipulable around the threshold, such as if villages attempted to influence census enumerators. This is quite unlikely given the rigorous Indian census process, and no other papers have ever found evidence of census data manipulation in India (e.g., Asher and Novosad 2020; Burlig and Preonas 2021). It is even less

plausible here since the delimitation process was not foreseeable at the time of the preceding census. Nonetheless, we test for continuity of the density of the village population distribution around the cutoffs (Cattaneo, Jansson, and Ma 2018), as seen visually in online Appendix Figure A3. The p -value is 0.44 for the 1991 population distribution and 0.35 for the 2011 distribution, indicating a lack of manipulation in both cases.

A second assumption is continuity across the regression discontinuity threshold of all village-level covariates that may be related to the outcomes. While this assumption can never be proven, we test this for observable baseline variables. Columns 2–7 of Table 1 examine variables from the relevant census round preceding delimitation (panel A for the 1995 delimitation with 1991 census data and panel B for the 2015 delimitation with data from the 2011 census) but find no evidence that any vary discontinuously across the population cutoff (see online Appendix Figures A4 and A5 for the corresponding figures).

The most likely reason why other variables would vary discontinuously across this population threshold is if there are other government programs that use the same threshold to determine eligibility. While we are unaware of any other programs that used population thresholds of 1,000 persons in the 1991 and 2011 censuses, we implement numerous robustness checks that indicate this cannot explain our results. We summarize them here but provide the details in Section IVE and online Appendix C. The first addresses the concern that there might be all-India programs that use the same threshold. We show that in other Indian states, the population of the GP that a village is assigned to does not change discontinuously at a village population of 1,000. We then use the same outcome data as our main analysis for Uttar Pradesh to test for discontinuities in these outcomes around that cutoff for other Indian states. The lack of discontinuities in those outcomes in the other states argues against national programs that target a population of 1,000.

While this rules out national programs, the remaining checks rule out programs specific to Uttar Pradesh that make use of the same cutoff. The second check leverages an implementation failure in which approximately 15 percent of Uttar Pradesh districts did not adhere to the delimitation rules. We test for discontinuities in the outcomes of interest in those districts. Third, we test for discontinuities in public goods provision around a village population of 1,000 in the 2001 census. The 2001 census round was not used for delimitation in Uttar Pradesh, but if the government commonly uses a village population of 1,000 for program targeting, we may observe jumps in public goods provision there. As a fourth check, we leverage the fact that at the time of the 2015 delimitation, some villages had populations around the cutoff but could not be split off; this was either because they were already the only village in their GP or their existing GP had a population below 2,000, so the other villages in their GP could not be split off into a sufficiently populous independent GP. These villages will be unaffected by the delimitation in 2015 but *would* be affected by any state or national program using the 1,000-person population cutoff for eligibility.

The final check, which we label as “alternative RD,” leverages how the delimitation of one village affects other villages. Suppose that between 1995 and 2014, villages a and b were in a GP with village c , but village c ’s population is just above 1,000 in the 2011 census. Village c will be split into a GP of its own. Villages a and b almost always remain together in their own independent GP, which now has a smaller population. However, villages a and b need not have populations near the

TABLE 2—EFFECTS ON VILLAGE AMENITIES

		2001		2011			2019
	Index	Primary school	Middle school	Primary school	Middle school	Above primary ed.	Education score
Panel A. Educational outcomes							
RD_Estimate	0.089 (0.022) [0.000]	0.027 (0.012) [0.024]	0.017 (0.010) [0.071]	0.019 (0.012) [0.110]	0.047 (0.015) [0.001]	13.739 (8.048) [0.088]	1.672 (0.436) [0.000]
Dep var mean	−0.110	0.834	0.122	0.846	0.427	574.208	31.257
Bandwidth	207	194	376	213	281	362	391
Effective observations	16,340	15,352	30,607	16,894	22,416	28,316	31,633
		2001		2011			
		Paved road	Tap water	Paved road	FPS (= 1)	Community center	
Panel B. Village-level infrastructure							
RD_Estimate	0.052 (0.020) [0.008]	0.031 (0.012) [0.008]	0.015 (0.014) [0.266]	−0.007 (0.011) [0.541]	0.027 (0.012) [0.030]	0.020 (0.013) [0.110]	
Dep var mean	0.265	0.561	0.391	0.685	0.770	0.325	
Bandwidth	331	436	321	417	228	340	
Effective observations	26,717	35,848	25,094	34,151	18,128	27,475	
		House		Sanitation			
		House materials	Piped water	Toilet	Open defec.	Closed drains	
Panel C. Household-level infrastructure							
RD_Estimate	0.092 (0.023) [0.000]	0.039 (0.016) [0.015]	0.038 (0.727) [0.958]	3.247 (0.651) [0.000]	−3.342 (0.679) [0.000]	1.199 (0.470) [0.011]	
Dep var mean	−0.044	0.087	20.319	17.795	81.053	7.005	
Bandwidth	243	304	367	218	216	254	
Effective observations	19,246	24,377	29,919	17,212	17,060	20,209	

Notes: This table reports regression discontinuity estimates from the main estimating equation of the effect of being above the treatment population threshold on different public service–related outcomes. Panel A presents outcomes related to education: whether the village had a primary or middle school present in 2001 or 2011 (as measured in the census), the number of residents with a primary education or above in 2012 (Socio-economic and Caste Census), and the education score of the village in the 2019 Mission Antyodaya survey. The outcomes in panel B are whether the village has an all-weather road in 2011, whether the village has tap water in 2001, whether the village has an all-weather road in 2011, whether the village has a fair price shop within the village in 2001, and whether the village has a community center in 2011 (all measured in the respective census round). The outcomes in panel C are an index of fraction of village residents living in houses made of high-quality materials in 2011, having in-house piped water in 2011, having a toilet in 2011, primarily defecating in the open in 2011, and with closed-drain sanitation systems. See the online Appendix for details of each dataset. The running variable is the population of the village in the 1991 census. Each specification uses a linear polynomial, triangular kernel, and MSE-optimal bandwidth estimated following Calonico et al. (2017). Standard errors are clustered at the gram panchayat level and reported below the point estimates. *p*-values are reported within brackets.

1,000-person threshold, so if we observe treatment effects for villages *a* and *b*, these cannot be the result of another state-specific program using the 1,000-person cutoff. The results of all of these tests are discussed in detail in Section IVE and indicate that this threshold was not used for other programs.

IV. Results

A. Village-Level Infrastructure

We begin by examining educational outcomes in panel A of Table 2, using a combination of data from the 2001 and 2011 censuses (presence of primary/middle/secondary/senior secondary schools in both rounds), the 2011–2012 Socioeconomic and Caste Census (educational attainment of residents), and the 2019 Mission Antyodaya survey (educational facilities score). Since this is a large number of outcomes, we create a standardized index following Kling, Liebman, and Katz (2007) to deal with concerns about multiple hypothesis testing.¹³ Column 1 runs the main regression discontinuity specification for the education index and finds that it increases by 0.089 standard deviations across the threshold, where this increase is highly statistically significant ($p < 0.001$; see panel A of Figure 2 for the corresponding figure).

The remaining columns of Table 2 show effects on each variable of the index to understand where the effects are coming from. In the short run, there is significantly more educational infrastructure present in villages to the right of the cutoff, which are 2.7 percentage points more likely to have a primary school (3.2 percent) and 1.7 percentage points (13.9 percent) more likely to have a middle school in 2001. This difference expands in the medium run, as those villages become 4.7 percentage points (11.0 percent) more likely to have a middle school by 2011, although the coefficient on primary schools (1.9 percentage points) is no longer statistically significant at conventional levels ($p = 0.11$).¹⁴

The Mission Antyodaya survey provides a longer-run perspective. In 2019, Mission Antyodaya collected village-level data on educational infrastructure, such as schools and facilities within the schools (toilets, midday meal, etc.), and created an education score for each village, which we rescale to be on a 0–100 scale. The value of this score is 1.67 points (0.095 standard deviations) higher in villages to the right of the threshold, implying that the effects of polity size are persistent even 24 years after the original delimitation.¹⁵ A nice feature of our setting is that we observe both the short and long run, so would capture dynamics that take longer to develop, such as catch-up.

Khanna (2023) shows that provision of this type of basic educational infrastructure over a similar time period improves educational attainment and earnings in India, so we would expect the same here. We also test for this directly using village-level data from the 2011–2012 Socioeconomic and Caste Census. Among other outcomes, the SECC measures completed education by every member of the

¹³The indexes in Kling, Liebman, and Katz (2007) are calculated by generating an equally weighted average of the z-scores of each variable in the index. To calculate the z-score for each variable, we subtract the mean of the variable among villages just below the cutoff and then divide by the standard deviation among that group.

¹⁴The introduction of the Sarva Shiksha Abhiyan program in November 2000, a federal education program, may have reduced the gap in primary schools between the treated and untreated villages. The results for the presence of secondary and senior secondary schools are reported in online Appendix Table A3.

¹⁵This is measured four years after the redelimitation in 2015, at which point virtually all of the villages (> 97 percent) with populations just below the discontinuity in 1995 now have populations above it. The estimated difference can thus be interpreted as reflecting an accumulated stock of benefits from smaller polity size over the 1995 to 2014 period. That may be a lower bound on long-run effects since villages that were below the discontinuity in 1995 but were allocated to a smaller polity in 2015 may have experienced some catch-up between 2015 and 2019.

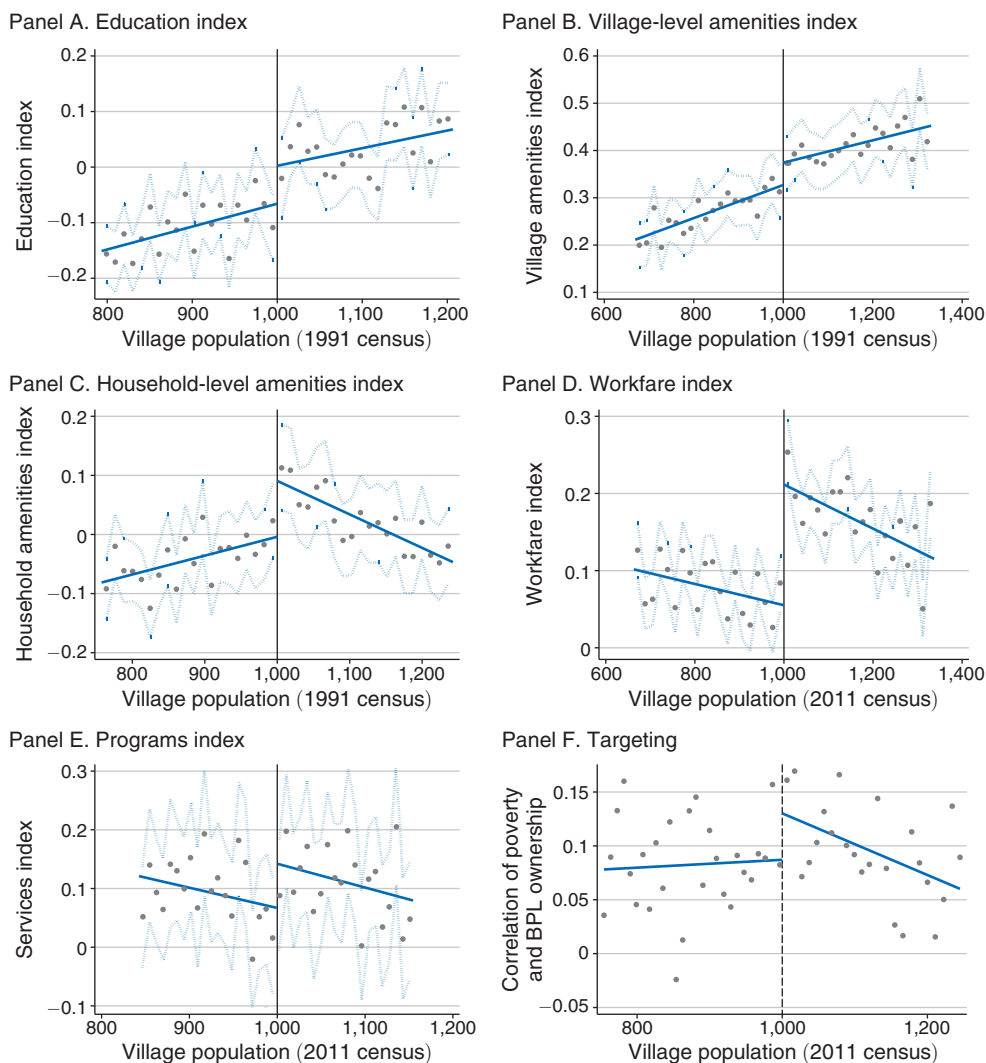


FIGURE 2. EFFECT ON MAIN OUTCOMES

Notes: This figure plots the relationship between indexes of various outcomes and the village's population at the time of the relevant delimitation episode. Points to the right of 1,000 are above the treatment threshold, while points to the left are below the threshold. Each point represents approximately 480 villages, and the bandwidth is based on the optimal bandwidth selection from Cattaneo, Jansson, and Ma (2020). Panel A plots the relationship between an index of educational outcomes and village population in 1991. The outcomes composing the index include the existence of primary, middle, secondary, and senior secondary schools in 2001 and 2011, educational attainment in 2012, and educational infrastructure score from the Mission Antyodaya data in 2019. Panel B plots this for an index of measures of village-level infrastructure, while panel C does this for household-level infrastructure. The village-level infrastructure includes the presence of a paved road in 2001 and 2011, piped water in 2001, and the presence of a fair price shop and community center in 2011. The household-level infrastructure index is based on the fraction of residents living in a house made of high-quality materials, having in-house piped water, having a toilet, and having a closed drainage system for sanitation in 2011. Panel D examines an index of outcomes related to the National Rural Employment Guarantee Scheme workfare program, while panel E studies an index of the delivery of eight individually targeted programs. Online Appendix Figures A6–A9 provide plots for each of the individual outcomes that make up the indexes. Panel F plots a binscatter of the relationship between village population and the correlation of poverty and ownership of BPL ration cards within village population bins of five.

village. Consistent with Khanna (2023), there is an increase in completion of primary education. The point estimate is economically large considering that only residents of school-going age over the ten or so years prior to 2012 could be affected by this construction. We take this as evidence that the additional school construction affects educational attainment; these effects on educational attainment will plausibly increase over time, as a larger fraction of the village is of such an age that they could have attended school when the additional facilities were in place.

Panel B of Table 2 examines other forms of village-level infrastructure that are the responsibility of the gram panchayat: the presence of all-weather roads, availability of tapped water in the village, and presence of a fair price shop or community center.¹⁶ The effects on an index of those variables are less strong than for educational infrastructure, with an effect of 0.052 standard deviations on the Kling, Liebman, and Katz (2007) index, but are still statistically significant at the 1 percent level ($p = 0.008$). In both 2001 and 2011, the census measures whether the village has an all-weather approach road made out of gravel/tar. Treated villages initially appear to have higher provision of all-weather roads (3.1 percentage points) in 2001, but this disappears by 2011.¹⁷ This could be due to catch-up from large national road-building programs that occurred after 2001 (Asher and Novosad 2020). We do not observe effects on whether tap water is available in the village in 2001, but villages to the right of the discontinuity are more likely to have their own fair price shop (FPS), which provides subsidized grains to poor households. The coefficient on the presence of a community center is positive but just short of statistical significance at conventional levels ($p = 0.11$).

These results demonstrate that smaller GPs provide more infrastructure and that the effect is fairly persistent over time.¹⁸ Given that decisions on the allocation of these types of infrastructure are primarily made at higher levels of government, this at least partially reflects a higher level of effort by GP leaders in lobbying politicians and those officials. It may also reflect reductions in corruption, where infrastructure projects in India, and in Uttar Pradesh in particular, are notorious for high levels of corruption (Wilkinson 2006; Lewis-Faupel et al. 2016).¹⁹ Corruption affects both whether infrastructure such as schools are ever built as well as whether the construction is of sufficient quality that it remains functional. A nice feature of our outcome

¹⁶Table 2 measures effects on village infrastructure outcomes that are available in our data and for which responsibility has actually been devolved to GPs in Uttar Pradesh (see Tables 11 and 12 of Indian Institute of Public Administration 2020 for a list of devolved functionalities). Panels B and C of online Appendix Table A3 use the main RD approach to investigate census outcomes that have been devolved on paper but not in practice (Indian Institute of Public Administration 2020). Given that these functionalities have not been devolved in practice, it is unsurprising that there is little effect on most of these outcomes.

¹⁷Asher and Novosad (2020) study an Indian road construction program based on population cutoffs, but those were based on the 2001 census population rather than 1991. Furthermore, Asher and Novosad (2020) find that Uttar Pradesh did not follow population cutoff rules for road construction.

¹⁸One factor cited in favor of larger polities is that larger units may internalize negative externalities and so may make more socially optimal decisions (Lipscomb and Mobarak 2016). While most of the outcomes we can observe are not characterized by negative externalities, this is true for one: irrigation, where greater irrigation in one village can lead to less in neighbors due to competition over canal water or groundwater. Panel B, column 3 of online Appendix Table A3 tests this by looking how the fraction of land in the village that is irrigated varies across the discontinuity. We do not observe an effect on the total amount of land that is irrigated, though it could be that the negative spillovers are sufficiently long-term or small that we cannot detect them.

¹⁹See Wilkinson (2006) for illustrative anecdotes of the corruption of local politicians in Uttar Pradesh infrastructure spending. These include a \$150 million program in which audits found that a third of funds disappeared and only 44 percent of the program targets were achieved.

variables is that they measure on-the-ground presence rather than stated allocations and so capture both lobbying and corruption, even if we cannot disentangle the two.²⁰

B. Individually Targeted Benefit Programs

In addition to village-level public goods, local governments play a role in many individually targeted benefit programs. These fall under the second category from Section IC: programs that are initiated by individual applications or at the village level but are funded by higher levels of government. A large literature has documented how GP leaders often provide information to citizens on how to access these programs, assist with preparing applications, and interface with bureaucrats to aid citizens whose applications are held up (Chauchard 2017; Bussell 2019; Kruks-Wisner 2018). Polity size may change leader incentives in ways that affect how they carry out those roles.

We first examine aspects of household physical infrastructure that are potentially affected by government programs: whether a house is constructed from high-quality materials like brick, has piped water, has a toilet, and is connected to closed drains. We use data from the 2011 census, which reports the fraction of households in a village possessing each of these characteristics. We construct an index of these variables, and using the RD, find improvements of 0.092 standard deviations in smaller polities ($p < 0.001$; panel C of Table 2).

The first component of the index is the quality of materials from which houses are built. Polity size could plausibly affect this through channels outside of the implementation of social welfare programs (e.g., general economic growth), but a particularly likely channel is state and national government programs to assist with housing. The largest of these is the Indira Awas Yojana (IAY, renamed to “Pradhan Mantri Gramin Awas Yojana” in 2015), a national program that provides the rural poor with funds to upgrade the quality of their homes. Since 1985, over 25 million households have been assisted by this program, and political scientists have documented a prominent role of local government leadership in helping citizens access it (Dunning and Nilekani 2013; Bussell 2019). We use data from the 2011 census that measures the fraction of house roofs, walls, and floors in a village that are made out of particular materials (e.g., brick, grass). We code each material as high/low quality and calculate the total fraction of roofs/walls/floors built of high-quality materials.²¹ We then combine these three variables into a Kling, Liebman, and Katz (2007) index to measure the quality of village housing stock. Panel C, column 2 of Table 2 finds that villages to the right of the discontinuity have houses built of higher-quality materials. At the same time, we do not detect effects on whether the household has piped water (column 3), perhaps reflecting fewer government programs dedicated to piped water provision before 2011.

²⁰ A commonly cited benefit of larger polities is benefiting from economies of scale, and these forms of physical infrastructure are the types of high-fixed cost items for which economies of scale seem most plausible. However, since the geographical distribution of the population remains the same regardless of how citizens are divided into polities, there are not standard economies of scale in polity size for the types of goods that we investigate here. See online Appendix G for a more detailed explanation.

²¹ We code a material as high quality if there is a positive correlation in the census data between the fraction of houses in a village constructed from that material and the fraction that census enumerators reported were in good condition. Online Appendix Table A4 examines roofs/walls/floors separately as well as effects on the individual materials used. Most of the effect is explained by more houses being constructed from brick.

Another program in which GP leadership played an important role was the Total Sanitation Campaign (TSC). This began in 1999 and encouraged construction of toilets with information campaigns and cash transfers to cover 80 percent of toilet construction costs, eventually funding the building of 1 toilet per 10 people in rural India (Spears and Lamba 2016). TSC relied heavily on GP leaders for implementation; a 2013 survey of 700 Uttar Pradesh households who had received toilets through TSC found that 99.4 percent received support from their GP leaders (Planning Commission of India 2013). The 2011 census measures the percent of households in each village who own a toilet, which we use as a measure of successful TSC implementation. Panel C, column 4 of Table 2 finds that toilet ownership jumps by 3.2 percentage points around the discontinuity, reducing open defecation by roughly the same amount (column 5). Since funding came entirely from the state and national government rather than the GP budget, these results would reflect effort by GP-level politicians in assisting constituents in accessing the program.²²

We also find that 1.2 percentage points more households were connected to closed-drain sanitation systems, as opposed to uncovered drains or no drainage system at all. This may be related to the Total Sanitation Campaign or the use of GP budgets to serve the expanded toilet network. But given the serious consequences of poor sanitation environments for children's health, these sanitation improvements likely have meaningful public health benefits (Spears and Lamba 2016; Geruso and Spears 2018). Note also that since the household infrastructure outcomes reflect successfully completed projects, our estimates incorporate any diversion of spending as well as the quality of implementation, where shoddy construction will be more likely to fall apart by the time of the census.

Next, we study the implementation of a series of public benefit programs. We focus on data from Mission Antyodaya that measure the number of beneficiaries from eight government programs in each village of Uttar Pradesh in 2019. This is based on administrative records from the relevant government departments, where the programs are (i) BPL ration cards, which entitle the holders to purchase subsidized grains from government ration shops; (ii) publicly provided health insurance (Pradhan Mantri Jan Arogya Yojana); (iii) pensions for the elderly, widows, and the disabled (National Social Assistance Programme); (iv) household electricity connections through the Saubhagya scheme (Pradhan Mantri Sahaj Bijli Har Ghar Yojana); (v) receipt of a liquified petroleum gas (LPG) connection (Pradhan Mantri Ujjwala scheme); (vi) receipt of a Rs5,000 cash transfer for pregnant women and mothers (Pradhan Mantri Matru Vandana Yojana, or PMMVY) scheme; (vii) receiving housing subsidies or being on the waiting list for subsidies (either the Pradhan Mantri Awaas Yojana or state-specific schemes); and (viii) having a zero-balance bank account (Jan Dhan initiative).

Since these program data are from the post-2015 period, we run the main RD specification using the 2015 delimitation episode with 2011 census population as

²²This and the housing materials analysis measure end user outcomes. Online Appendix Table A5 supplements this with data from the Ministry of Housing and Urban Affairs, measuring the construction of IAY homes in 2014–2015, and from the 2011 census, recording whether a village participated in the Total Sanitation Campaign. Consistent with IAY and TSC playing a role in housing/toilet improvements, villages to the right of the discontinuity are more likely to construct IAY houses and participate in TSC. Panel A, column 7 of Table 3 similarly finds that smaller polities in 2019 have better access to housing programs.

TABLE 3—EFFECT ON DELIVERY OF SERVICES

	Programs index	BPL card	Health insur.	Pension	Saubhagya	LPG	Housing benefits	PMMVY	Jan Dhan
<i>Panel A. Individual-level programs</i>									
RD_Estimate	0.088 (0.033) [0.009]	5.016 (2.420) [0.038]	2.210 (1.059) [0.037]	−1.346 (1.075) [0.210]	2.550 (1.396) [0.068]	5.627 (2.451) [0.022]	5.538 (2.186) [0.011]	0.099 (0.134) [0.460]	4.974 (2.048) [0.015]
Dep var mean	0.096	92.962	22.322	35.241	42.110	96.017	41.564	2.575	69.316
Bandwidth	156	316	275	283	378	209	203	365	213
Effective obs	12,594	25,213	22,011	22,572	30,087	16,844	16,234	29,000	17,079
	NREGS index	Work demand	Days worked	Labor expend.	Material expend.	Total projects			
<i>Panel B. Workfare program implementation</i>									
RD_Estimate	0.162 (0.026) [0.000]	0.011 (0.001) [0.000]	0.308 (0.049) [0.000]	63.633 (9.445) [0.000]	22.864 (5.542) [0.000]	0.006 (0.001) [0.000]			
Dep var mean	0.079	0.078	2.039	394.114	147.997	0.086			
Bandwidth	337	373	347	338	308	334			
Effective obs	119,034	131,803	122,152	119,284	108,397	117,724			

Notes: This table reports regression discontinuity estimates from the main estimating equation of the effect of being above the treatment population threshold on different welfare program and NREGS outcomes. Panel A presents outcomes related to welfare programs as measured in the Mission Antyodaya data in 2019. Column 1 is an index of the eight programs calculated following Kling, Katz, and Liebman (2007). The outcomes are the number of beneficiaries of (i) BPL ration cards, which entitle the holders to purchase subsidized grains from government ration shops; (ii) publicly provided health insurance through the Pradhan Mantri Jan Arogya Yojana; (iii) pensions for the elderly, widows, and the disabled under the National Social Assistance Programme; (iv) household electricity connections through the Saubhagya scheme (Pradhan Mantri Sahaj Bijli Har Ghar Yojana); (v) receipt of a liquified petroleum gas (LPG) connection through the Pradhan Mantri Ujjwala scheme; (vi) receipt of a Rs5,000 cash transfer for pregnant women and mothers through the Pradhan Mantri Matru Vandana Yojana (PMMVY) scheme; (vii) receiving housing subsidies (or being on the waiting list for subsidies) through the Pradhan Mantri Awaas Yojana or state-specific schemes; and (viii) having a zero-balance bank account (Pradhan Mantri Jan Dhan initiative). In panel B, the first column is an index of the other NREGS outcomes. The remaining columns are the proportion demanding work through NREGS, the days worked per person, the total expenditure on wages for labor, the total expenditure on materials, and total number of NREGS projects completed. See the online Appendix for details of each variable. The running variable is the population of the village in the 2011 census. Each specification uses a linear polynomial, triangular kernel, and MSE-optimal bandwidth estimated following Calonico et al. (2017). Standard errors are clustered at the gram panchayat level and reported below the point estimates. *p*-values are reported within brackets.

the running variable. As this is a large set of outcomes, we combine them into a single index following the method of Kling, Liebman, and Katz (2007). Panel A, column 1 of Table 3 finds that villages to the right of the discontinuity have a 0.088 standard deviation higher value of this index (see panel E of Figure 2 for the corresponding RD plot). The remaining columns of the table conduct this analysis for each program individually, finding improvements across most programs rather than a single program driving the result.^{23, 24} These results are informative about a

²³Online Appendix Table A6 finds that the improvements documented in this section do not translate into higher incomes, using data from the 2011–2012 Socioeconomic and Caste Census. This is not terribly surprising given that many of our outcomes are related to access to government social support programs that are unlikely to substantially improve income generation even if they do improve welfare. Although more education should increase earnings eventually (Khanna 2023), there is not sufficient time for the children benefiting from this to age into the labor market in sufficient numbers to affect aggregate economic activity by 2012.

²⁴A challenge with the 2015 delimitation is that villages to the left of the discontinuity could still be affected by splits. Suppose villages *X*, *Y*, and *Z* are in the same GP prior to 2015, where *X* has a population of 1,001, and *Y*

different dimension of government performance than the infrastructure outcomes in Section IVA; these indicate better performance of leadership in helping constituents make claims on the state (Kruks-Wisner 2018), where we will look deeper at possible reasons for this in Section V.

Targeting.—An extensive literature in economics has studied what factors influence the targeting of benefits (e.g., Coady, Grosh, and Hoddinott 2004; Alatas et al. 2012). In order to examine the extent to which these increases in public service delivery are well targeted, we focus on an outcome for which targeting criterion is well defined: possession of a BPL ration card, which entitles holders to highly subsidized grains. These are intended to go to poor households, but other work has shown that many do not (Gulati and Saini 2015).

We examine targeting with data from Asher et al. (2021), which estimates the fraction of households in every village that are below the poverty line using individual-level data from the 2011–2012 Socioeconomic and Caste Census. If targeting is good, we would expect a positive correlation between the fraction of households in a village who hold a BPL card (measured in the Mission Antyodaya data) and the poverty rate of the village (as measured in the SECC by Asher et al. 2021). If smaller polities have better targeting, this poverty rate-BPL correlation may increase around the discontinuity. Panel F of Figure 2 visualizes this by calculating the correlation between BPL card ownership and poverty across villages within bins of 5 (e.g., correlation within villages with populations of 750–754, 755–759, etc.). The correlation appears to be higher in villages to the right of the population discontinuity, consistent with better targeting. We use the corresponding regression specification to conduct a formal statistical test:

$$\begin{aligned}
 (2) \quad BPL = & \beta_0 + \beta_1 poverty + \beta_2 poverty \times population \\
 & + \beta_3 poverty \times population \times above + \beta_4 poverty \times above \\
 & + \beta_5 population + \beta_6 population \times above + \beta_7 above + u,
 \end{aligned}$$

where *poverty* is the poverty rate of the village as measured in the SECC, *population* is the village's population in the 2011 census, and *above* is a dummy variable equal to 1 if that population is above 1,000. β_1 measures the relationship between poverty and BPL card ownership, the second term allows this relationship to vary with the village population, and the third term allows this relationship to vary above and below the discontinuity. The coefficient of interest is β_4 , which measures whether the relationship between poverty and BPL card ownership becomes discontinuously stronger at the threshold. The remaining terms saturate the model and correspond to a standard RD equation in population; β_7 can be interpreted as the change in BPL cards at the discontinuity if the poverty rate of the village is equal to zero. Online

and Z both have populations of 999; Y and Z would end up together in a smaller GP without X. This would plausibly make our estimates conservative by weakening the first stage of how the discontinuity affects GP population. When we rerun this analysis in online Appendix Table A7 but drop these potentially contaminated villages, estimates are nearly identical (for example, the point estimate in column 1 changes from 0.088 to 0.083) so this does not appear to be a major issue.

Appendix Table A8 reports estimates of β_4 and β_7 . Consistent with the figure, the estimate of β_4 indicates that the relationship between poverty and BPL card ownership increases discontinuously at the threshold ($p = 0.019$), consistent with better targeting as a result of smaller polity size. These improvements could reflect greater familiarity with community members, more egalitarian dissemination of information about programs, or more assistance to poor households in program applications, although we cannot disentangle between these or other possibilities with our data.

C. Workfare (*National Rural Employment Guarantee Scheme*)

Finally, we examine the National Rural Employment Guarantee Scheme workfare program. We use administrative data over the period from 2016 to 2020, with outcomes including the number of NREGS projects completed, reported demand for work, days of work provided, and expenditure on labor and materials in each year. Nearly all GPs (99 percent) have at least 1 NREGS project per year, and approximately 15 percent of households have at least 1 member working in NREGS in a given year (1.21 workers per household). We normalize all of the outcomes by dividing by total residents for ease of interpretation (e.g., NREGS wages per resident).²⁵

Since each of the NREGS outcomes are highly correlated with each other, we focus on an index created following Kling, Liebman, and Katz (2007). In panel B of Table 3 and panel D of Figure 2, we find that the index is 0.16 standard deviations higher in villages to the right of the discontinuity ($p < 0.001$). The remaining columns of Table 3 show that improvements are consistent across each of the outcomes. However, one concern is that the changes in these or other services may differ throughout the distribution of households; for example, gains could be concentrated among advantaged groups if smaller polities are more subject to elite capture. A nice feature of these data is that there are measures of *who* within the community benefits. For three of the variables (number of households possessing job cards to work for NREGS, total person-days worked, and proportion of households who received NREGS work), the data break down the outcome by whether the beneficiaries are from a scheduled caste (SC), and for person-days worked, it is additionally broken down by gender. Panel A of online Appendix Table A9 reruns the main RD specification with these outcomes (columns 1, 3, and 5 for SC, columns 2, 4, and 6 for non-SC) and finds similar gains for both SC and non-SC workers. Gains for female workers are smaller in magnitude (column 7), reflecting the lower rate of NREGS work among women in Uttar Pradesh, but the increase is similar in proportional terms. These findings suggest gains from smaller polities are shared throughout the community.

The workfare outcomes reflect how GP leaders manage social programs at the village level, where they propose projects, hire workers, deal with higher-level administrators for funding, and manage project implementation. As with the earlier outcomes, the NREGS measures reflect leader effort but may also reflect managerial competence. Section V will dig deeper into what explains these findings.

²⁵The data are at the GP level, so this standardizes the outcomes. This denominator includes all residents, including those who do not work in NREGS. See online Appendix H for details.

D. Estimating Heterogeneity by Population Size

The previous analysis found consistent evidence of increased public service provision in smaller polities. This approach estimated the treatment effect of allocation into a GP with a population of approximately 1,000 (the cutoff value) instead of a GP with a larger counterfactual population.²⁶ However, the treatment effect may be heterogeneous. More precisely, there may be a treatment effect $\tau(x, y)$ of shifting from a GP with a population of x residents into a GP of y residents, where the effect depends on x and y . For example, the effect of moving a village from a GP of 6,000 people to one of 1,000 (i.e., $\tau(6000, 1000)$) could be larger than moving from a GP of 3,000 to one of 1,000 (i.e., $\tau(3000, 1000)$); if so, splitting up larger GPs may be most beneficial. Following this notation, the previous sections estimated $\int_x w(x)\tau(x, 1000)dx$, where x refers to the GP population if the village were not split off and the density function $w(x)$ reflects the prevalence of this value x among villages near the discontinuity. While that is a policy-relevant estimate of the effect of Uttar Pradesh's delimitation policy, a policymaker would prefer to know $\tau(x, y)$ to determine polity boundaries.

We use two estimation strategies to understand $\tau(x, y)$. Our first approach is to estimate $\tau(x, 1000)$ for the values of x that are observable in this context. This is possible in the 2015 delimitation because villages will typically remain with the same GP after 2015 unless a village is split off due to having a population above 1,000. Thus, for the 2015 delimitation, a village's value of x is the population of its pre-2015 GP. We split the sample into bins of pre-delimitation populations and reestimate the main RD equation within each subsample.

Panel A of Figure 3 plots these estimates and 95 percent confidence intervals for the index of workfare outcomes, while panel B does this for the index of individual-level programs.²⁷ For both outcomes, the treatment effect $\tau(x, 1000)$ appears to increase with the value of x ; that is, the largest treatment effects are found in villages that otherwise would have been in very large GPs. We test this in columns 1 and 2 of Table 4 by running the standard RD but also including an interaction term with population. While the interaction with GP population is strongly statistically significant for workfare ($p < 0.001$), it is not for individual-level programs (although the coefficient is positive). We interpret this as demonstrating larger treatment effects from splitting up the largest polities for the set of workfare outcomes.²⁸

Our second approach estimates $\tau(x, x - 1000)$ for different values of x . For this, we leverage a second source of exogenous variation generated by the delimitation rules. If a village is above the population cutoff in the 2015 delimitation, it is supposed to be split off from its pre-2015 GP. This split also affects the other villages in its pre-2015 GP, where those villages will typically remain together in a GP that is now approximately 1,000 people smaller (e.g., for a GP that had a population of

²⁶ Panels E and F of Figure 1 help visualize the counterfactual population distribution; they reestimate the main RD equation with the dependent variable as a binary variable equal to one if the village's post-delimitation GP is within a particular population range and then plot the coefficients for different ranges. The figure indicates that the counterfactual GP population is typically between 2,000 to 3,500.

²⁷ Online Appendix Figure A11 plots these for each of the workfare outcomes separately. This analysis is not possible with the 1995 delimitation exercise because we don't observe pre-1995 GPs.

²⁸ Due to the smaller main effect on the programs index, it is unclear whether the non-statistically significant interaction term in column 2 results from low statistical power or a true lack of interaction with population.

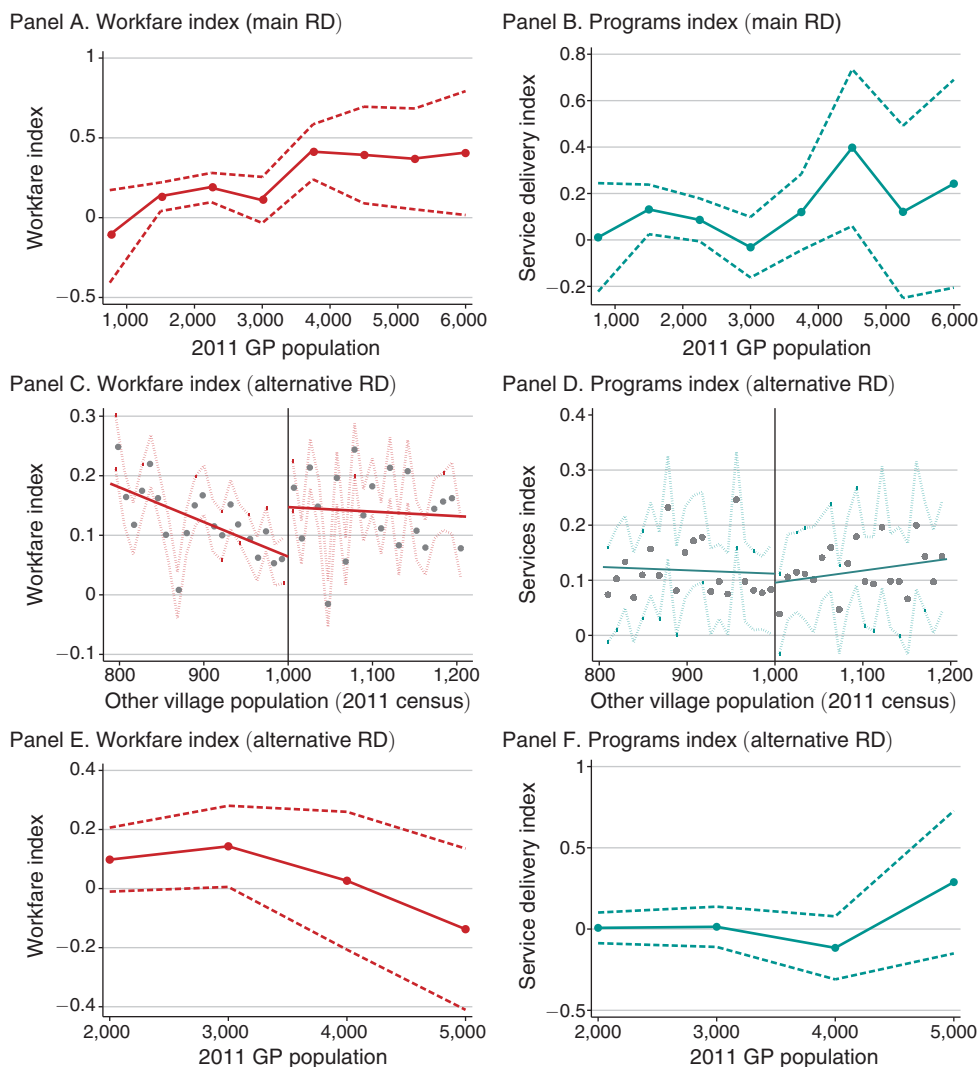


FIGURE 3. EFFECTS ON WORKFARE AND INDIVIDUAL-LEVEL SERVICE DELIVERY (2015 DISCONTINUITY)

Notes: This figure plots the relationship between outcomes related to services delivery and the village's population in the 2011 census. Panels A, C, and E examine indexes of outcomes related to the National Rural Employment Guarantee Scheme, while the other panels study an index of the delivery of eight individually targeted programs. In panels A and B, we divide the sample villages based on the population of their GP prior to the 2015 delimitation. We run the main RD specification within each of these bins to look at the effect of being above the treatment population threshold for villages within the bin. We then plot the coefficients and confidence intervals from each of these regressions in the figure. These estimates can be interpreted as telling us about the treatment effect of moving from that population range into a GP with a population of around 1,000. Panels C and D use an alternative RD approach based on the population of other villages in the same GP. These figures plot the RD graph for each of our main outcomes using this alternative RD. Panels E and F divide all of the villages in our sample based on the population of their GP prior to 2015 (2,000–2,999, ..., 5,000–5,999). We run the alternative RD specification within each of these bins; this measures the effect of having another village within your pre-2015 GP be above the treatment population threshold. We then plot the coefficients and confidence intervals from each of these regressions in the figure. These estimates can be interpreted as telling us about the treatment effect of moving from that population range into a GP with a population of approximately 1,000 fewer people (e.g., the treatment effect of moving from a GP of 3,000 people to one with 2,000).

TABLE 4—ESTIMATING HETEROGENEITY BY POPULATION SIZE (POST-2015 DELIMITATION)

	Village population RD		Other village population RD			
	NREGS index (1)	Programs index (2)	NREGS index (3)	NREGS index (4)	Programs index (5)	Programs index (6)
RD_Estimate	0.164 (0.030) [0.000]	0.095 (0.038) [0.012]	0.098 (0.042) [0.019]	0.131 (0.044) [0.003]	0.007 (0.035) [0.843]	−0.011 (0.038) [0.781]
RD_Estimate × 2011 GP population	0.467 (0.087) [0.000]	0.032 (0.114) [0.777]	—	−0.326 (0.119) [0.006]	—	0.121 (0.116) [0.296]
Bandwidth	353	196	192	192	230	230
Effective observations	79,282	10,126	66,499	66,499	17,207	17,207

Notes: The first two columns of this table report heterogeneity in the effect of polity size with respect to previous GP population. This augments the standard RD equation with an interaction of the dummy variable for being above the threshold with the population of GP to which a village was assigned prior to the 2015 delimitation process. To implement this, we use the method of Calonico et al. (2017) for bandwidth selection and then run the corresponding regression with that bandwidth, adding variables for GP population and the interaction of the RD estimate and GP population, i.e., $y_i = \beta_0 + \beta_1 v_i + \beta_2 v_i \times above_i + \beta_3 above_i + \beta_4 gppop_i + \beta_5 above_i \times gppop_i + \epsilon_i$, where v_i is the population of village i in the 2011 census, $above_i$ is a dummy variable equal to 1 if v_i is greater than 1,000, and $gppop_i$ is the population of the GP with which the village is associated prior to the 2015 delimitation exercise. To avoid issues with outlier GP populations when examining this continuous interaction, we restrict to villages whose GP population was below the ninety-ninth percentile. We also restrict to villages whose prior GP population was above 2,000 since there is no first stage below that value (i.e., splits would not be allowed because that would create a GP with fewer than 1,000 people). Columns 1 and 2 use the main regression discontinuity from equation (1). Columns 3 to 6 use the alternative RD. Finally, for ease of interpretation of the interaction coefficient, we standardize the 2011 GP population variable so that it runs from 0 (GP population of 2000 in 2011) to 1 (GP population at the ninety-ninth percentile in 2011). For each gram panchayat in existence between 2010–2015, we determine the village whose 2011 census population was closest to 1,000. We drop those villages from the sample and use their population as the running variable for the other villages in their GP for the 2015 delimitation. Columns 3 and 5 show the main effects for the workfare and programs indexes. Columns 4 and 6 include the interaction from columns 1 and 2 to test for heterogeneity. Standard errors are clustered at the gram panchayat level. p -values are reported within brackets.

5,000 before the 2015 delimitation, the loss of a village that is just above the cutoff will reduce the GP population to approximately 4,000).²⁹ We can thus use the population of other villages in the same pre-2015 GP as a running variable in an alternative regression discontinuity design. For each gram panchayat in existence between 2010 and 2015, we determine the village whose 2011 census population was closest to 1,000. We drop those villages from the sample and use their population as the running variable for the other villages in their GP.³⁰ Online Appendix Figure A10 shows a strong first stage: being in a pre-2015 GP with a village whose population was just above rather than just below the cutoff results in being in a post-2015 GP with an average of 431 fewer people.

Column 3 of Table 4 reruns the analysis for the workfare index using this alternative RD approach (see panel C of Figure 3 for the corresponding figure) and finds an improvement of 0.098 standard deviations ($p = 0.019$). This is smaller than the

²⁹Note also that estimates from this method can be combined to trace out $\tau(x, y)$; e.g., $\tau(x, x - 2000)$ can be approximated by the sum of $\tau(x, x - 1000)$ and $\tau(x - 1000, x - 2000)$ under relatively weak conditions.

³⁰For this analysis, we restrict the sample to villages in GPs with populations greater than 2,000 at the time of the 2015 delimitation. GPs are supposed to contain at least 1,000 people, and so it is not possible to split a GP with fewer than 2,000 people to create two GPs of an adequate size. Thus, there is no first-stage effect of the discontinuity on polity size for GPs with populations below 2,000, as seen in panel F of Figure 1.

estimate in Table 3 (0.162 SD), likely because the difference in polity size across the alternative RD threshold is much smaller; for the alternative RD, villages to the right of the threshold are in GPs with 431 fewer people than villages to the left of the threshold, which is around half of the difference for the main RD within this sample (902 people). When the alternative RD estimate is rescaled to reflect this, the two designs produce very similar estimates of the effect of polity size on workfare performance.

The alternative RD estimate for individually targeted programs (column 5 of Table 4) is statistically insignificant. However, this is not surprising given that the main RD estimate for this outcome in Table 3 is 0.088. Even if there were an effect on the programs index from the alternative RD that were half the main RD estimate (due to the smaller difference in polity size across the threshold under the alternative RD, as discussed in the previous paragraph), that would be too small to reliably detect given the magnitude of the standard errors ($SE = 0.035$). Since it is hard to interpret the individual-level program results due to these power concerns, we focus on workfare when using this approach.

The main goal of this section is to understand how the treatment effect varies for different polity size changes. For that, we divide the villages into subsamples based on their pre-2015 gram panchayat populations (2,000–3,000, 3,000–4,000, 4,000–5,000, 5,000–6,000) and estimate the alternative RD within each subsample; this corresponds to estimating the treatment effect of moving from a GP of population 2,000–3,000 to one of 1,000–2,000, from 3,000–4,000 to 2,000–3,000, and so on. Panel E of Figure 3 plots the estimated coefficients when looking at the index of workfare outcomes. The effects for values of x between 2,000 and 3,999 are positive and statistically significant, but for values of x above 4,000, the effects are insignificant. While we can reject the equivalence of the effects for the first two subsamples as compared to the fourth subsample ($p = 0.095$, $p = 0.072$), we cannot reject equivalence between the effects for the third subsample and any of the others. Visually, it appears that there may be nonlinearities: while the estimates for the first two subsamples are positive and quite similar, the other two estimates are indistinguishable from zero. However, we lack statistical power to say this definitively. For a better-powered test, column 4 of Table 4 tests for whether the treatment effect varies linearly with 2011 GP population.³¹ Consistent with panel E of Figure 3, the interaction term is statistically significant ($p = 0.006$), where the treatment effect is stronger for smaller values of x .³²

The analyses in this subsection paint a more complex picture of the effects of polity size. The first analysis showed that the effects were largest when shifting a village out of a larger GP. However, the second analysis complicates the picture, where the effects were larger for smaller values of x . It could be that the gains are not simply from getting smaller but potentially about creating sufficiently small local governments, where above a certain threshold, allocation into smaller units does not have much of an effect.³³ We are limited by statistical power, particularly

³¹To avoid issues with outliers, we restrict the sample to GP populations below the ninety-ninth percentile. We also restrict to villages whose prior GP population was above 2,000 since there is no first stage on polity size below that value. To make the coefficients easier to interpret, we rescale pre-2015 GP population to range from 0 and 1, where 0 corresponds to 2,000 and 1 to the ninety-ninth percentile (8,191).

³²Column 6 of Table 4 and panel F of Figure 3 do not detect any effect for the individual-level programs, but this is unsurprising given the statistical power issues highlighted in the previous paragraph.

³³The changes are also bigger proportionally when x is smaller for the alternative RD, and so it may be that changes as a proportion of the population are what matters.

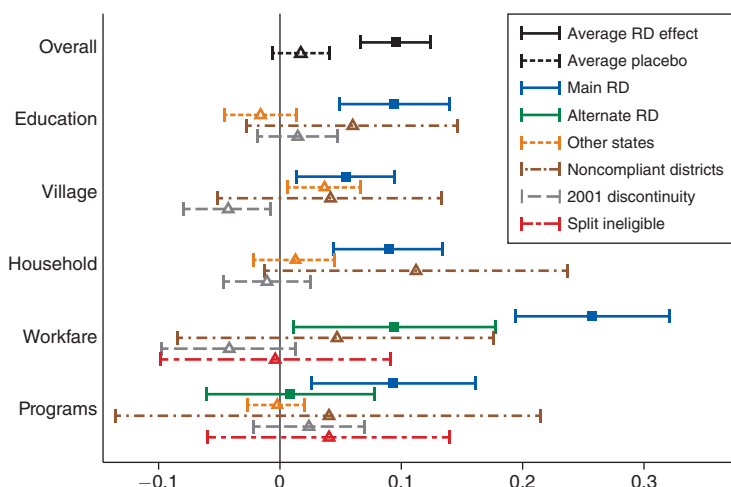


FIGURE 4. MAIN AND PLACEBO TESTS OF MAIN OUTCOMES OF INDEX

Notes: This figure plots regression discontinuity point estimates and 95 percent confidence intervals. These derive from the main estimating equation examining the effect of being above a particular population threshold on our five main categories of public service–related outcomes (education, village infrastructure, household infrastructure, workfare (NREGS), and services). Our main specification, examining the effect of polity size using own village population, is represented by the solid blue bar. The alternative RD estimate, which uses the population of another village within the same GP as a running variable, is the solid green bar. The other bars, which are denoted with dashed lines, correspond to placebo tests. The first test (orange) examines the main RD (using own village population around a cutoff of 1,000) in other states of India, which do not follow the same polity size rules. The second (brown) runs the main RD in a subset of Uttar Pradesh districts where the polity size rules were not followed, and the third (gray) uses own village population in 2001 as the running variable. The fourth (red) examines the RD for split ineligible villages. These are villages that in 2015 are in a GP with other villages whose population does not sum to more than 1,000. As a result, the other villages cannot form their own independent GP, so the focal village is ineligible to be split off even if it contains more than 1,000 people. For the first three outcomes, we cannot run the alternative RD or split ineligible placebo tests because those require knowledge of the prior GP composition, which we only have for the 2015 delimitation. We also cannot run the first placebo test (main RD in other states) for the NREGS outcomes since we only have NREGS data for Uttar Pradesh. The point estimates for the main RD and alternative RD in this figure are larger than those in Tables 2 and 3 because those tables include the villages from the noncompliant districts and split ineligible villages in the analysis sample. For the main and alternative RD specifications in this figure, the analysis sample only uses villages where the polity size rules are in effect. We also take the average of the nonplacebo and placebo coefficients, where we generate 95 percent confidence intervals by block bootstrapping at the 1995 GP level. In the bootstrap samples, we reestimate all of the 7 nonplacebo and 16 placebo coefficients and take the averages. We do this 500 times to generate confidence intervals.

in the analysis on individual-level programs, in making strong statements. However, if there is a threshold, it appears to be around 3,000 persons. Future research should delve more carefully into the possibility of such thresholds, particularly as additional outcome data become available in the next round of the Indian census and future rounds of delimitation expand the sample size for analysis.

E. Robustness Checks

This section discusses the robustness of our results to alternative explanations.

Placebo Checks.—First, as described in Section III, we test whether the government may use the same 1,000-person threshold for other programs. Figure 4 plots

the point estimates and 95 percent confidence intervals from these placebo tests for our key outcome variables (indexes for education, village infrastructure, household infrastructure, workfare, and individual-level programs). The placebo tests are denoted with dashed lines. The first test (orange) examines the discontinuity in other states of India, which do not follow the same polity size rules; the second (brown) runs the RD in a subset of Uttar Pradesh districts where the polity size rules were not followed; the third (gray) uses the 2001 village population as the running variable; and the fourth (red) examines the RD for split ineligible villages. Online Appendix C provides further details on each of the placebo tests. For comparison, we also plot the estimates from both the main and alternative RD approaches, which are denoted with solid lines.³⁴

Out of the 16 placebo tests, only 1 coefficient is positive and statistically significant at the 5 percent level (“village infrastructure” for other states). Given the number of placebo tests we ran, this is plausibly due to chance. To combine all the tests to account for multiple hypothesis testing, we construct 95 percent confidence intervals around the average of the placebo test coefficients using block bootstrapping; we cannot reject that on average there is no effect of the discontinuity in the placebo samples (see “Average placebo” in Figure 4; $p = 0.23$). Furthermore, the average of the placebo coefficient estimates is economically small (0.016). For comparison, we do the same thing for the actual tests of polity size using both the main and alternative RDs and find an average effect of 0.098 ($p < 0.001$). Six out of 7 actual tests of the effect of polity size return coefficients that are statistically significant at the 1 percent level, where Section IVD explains why the one non-statistically significant test is likely underpowered. We thus conclude that there is not strong evidence of use of the threshold by other programs.

Robustness to Other RD Specifications.—Next, we check robustness of the main tables of the paper to a different bandwidth selection method (CER-optimal), fixed bandwidths of 100 and 200, quadratic polynomials, and a uniform kernel. We include the key tables with a bandwidth of 200 in online Appendix Tables A10 and A11; the remainder are available in online Appendix J. The results are quite consistent with our main RD approach.

Spillovers.—A third concern is spillovers, and in particular, whether some of the measured benefits for treated villages come at the expense of untreated villages. Even if that were the case, our findings would still be informative in showing that smaller polities have greater public good access. However, the aggregate increase in overall public services would be smaller, and policy implications would be more ambiguous. Spillovers are most plausible onto other villages within the same block, as that is the next-highest level of government administration and where most of the competition for resources would occur.

³⁴ For the first three outcomes, we cannot run the alternative RD or split ineligible placebo tests because those require knowledge of the prior GP composition, which we only have for the 2015 delimitation. We also cannot run the workfare placebo test for other states of India since we only have NREGS data for Uttar Pradesh. The point estimates for the main RD and alternative RD displayed in this figure are larger than those in Tables 2 and 3 because those tables included the villages from the noncompliant districts and split ineligible villages in the analysis sample. We do not do that in Figure 4 since the whole point is to compare the placebo sample to the nonplacebo sample.

Online Appendix D leverages the delimitation rule to test for negative spillovers, measuring whether each block has a larger fraction of villages just above the threshold of 1,000 as opposed to just below. If there were negative spillovers, a village within a block that has a higher fraction of villages above the cutoff will have worse public services. We do not find evidence of that. One possible reason is if some of the effects come through reductions in corruption and so do not impose a fiscal cost. Another is that the extent of negative spillovers depends on whether there are hard budget caps and whether these caps bind in practice. If either condition does not hold, then more spending in one GP does not necessitate losses for another.³⁵ Since here only a small fraction of villages are delimited into a smaller polity, this can be accommodated by even limited budgetary slack. This may not hold with more extensive reforms to polity size, and so even though spillovers do not bias our estimates here, governments considering more intensive polity size reforms should think carefully about what this means in the context of their budgets.

Financing.—Finally, online Appendix E looks at whether our results are a product of fiscal funding formulas such that smaller local governments mechanically receive more funding on a per capita basis. In that case, even though our estimates would be internally valid for India, they would not necessarily generalize to contexts with different fiscal rules. The outcomes we study are funded from a mix of local government budgets and sources like district or block-level government departments. As noted in Section IC, local government budgets should be nearly equal on a per capita basis. We use data from the Panchayati Raj accounting system dating between 2013 and 2020 to confirm that there is no increase in per capita spending or tax revenues across the threshold. For the set of outcomes that are not funded by the local government budget, we test for funding formulas that favor smaller local governments, such as fixed per GP budgetary allocations that are invariant to population. We do not find any evidence for these possibilities, indicating that other mechanisms must explain these results, which we turn to in the next section of the paper.

V. Mechanisms

An outstanding question is what underlying mechanisms drive these findings. A better understanding of mechanisms can aid in assessing other contexts to which the estimates will generalize, as well as being informative as to the empirical relevance of theoretical mechanisms that have been proposed in the political decentralization literature. To structure our investigation, we divide up potential channels into four categories: political behavior of citizens, the characteristics of elected representatives, the incentives and work environment faced by representatives, and the composition of the polity. For each, we link to theories of decentralization relevant to that broad category. Our analysis related to the composition of the polity also helps disentangle the bundle of treatments that are necessarily part of reductions in polity size; smaller polities are simultaneously less populous, more geographically

³⁵ Instead, this would increase government debt and so must be paid down by future taxes; the question of these costs/benefits is discussed in online Appendix F.

compact, and more ethnically homogeneous, among other characteristics, but which of these treatments can account for the results?

A. Political Behavior of Citizens

Some of the most important ways through which polity size may affect government performance comes through political engagement. One possibility is that smaller polities may have higher levels of citizen engagement, as the cost of accessing leaders is lower and the likelihood of being a pivotal voter is higher. This increased engagement may put pressure on leaders to perform. Mookherjee (2015) notes “[t]he possibility that [political participation and competition] may be affected by decentralization in the long run has been largely ignored,” so we try to address this gap.

Panel A of Table 5 uses our main RD specification to examine how allocation into smaller GPs affects citizen engagement.³⁶ We begin by examining attendance at village meetings (*gram sabha*) in which citizens have the opportunity to influence decisions of their local government. We use data from the Ministry of the Panchayati Raj on meetings dedicated to creating an annual “Gram Panchayat Development Plan,” in which the local government decides on the development priorities for the upcoming year. The meetings are an important venue through which citizens can influence local government decisions and public goods provision (Rao and Sanyal 2010; Ban, Jha, and Rao 2012; Das 2015); Das (2015) estimates that a 1 standard deviation increase in the relative attendance of women to men at a gram sabha increases the provision of sanitation and health facilities by 0.5 and 0.6 standard deviations, respectively. We find that attendance is substantially higher in smaller polities, with a 2.1 percentage point increase (28 percent, $p < 0.01$) (panel A, column 1). The increases are similar as a percentage of the dependent variable mean for disadvantaged groups, where scheduled caste attendance goes up 21.5 percent (2.3 percentage points, $p < 0.01$) and female attendance increases by 25 percent (0.3 percentage points, $p < 0.01$).

Next, we examine other forms of political engagement using data from three rounds of gram panchayat elections (2010, 2015, and 2020). Panel A, column 4 finds that individuals are significantly more likely to turn out to vote in smaller GPs (1.9 percentage points). Column 5 examines whether citizens are more likely to stand as candidates for *pradhan* in small polities, a more personally costly form of civic engagement. Likelihood of running for office increases by around a quarter, with an increase in 1.12 candidates per 1,000 over a dependent variable mean of 4.85. Column 6 shows that estimates are similar if we restrict only to candidates who gain at least 5 percent of the vote share, indicating that this is not just about attracting marginal candidates.

Taken together, the evidence suggests that smaller local governments have higher rates of political engagement. Das (2015) provides direct causal evidence that attendance in gram sabhas affects the allocation of public goods, and work from other contexts indicates that high rates of voter turnout can translate into better public

³⁶In calculating all of the p -values in this table, we apply the Romano-Wolf correction, which controls the family-wise error rate.

TABLE 5—EFFECTS ON GP ELECTION OUTCOMES

	Gram sabha			Election		
	Total partic.	SC partic.	Female partic.	Voter turnout	Candidates per 1,000	Eff. candidates per 1,000
<i>Panel A. Political participation</i>						
RD_Estimate	0.021 (0.003) [0.002]	0.023 (0.005) [0.002]	0.003 (0.001) [0.002]	1.930 (0.262) [0.002]	1.119 (0.074) [0.002]	0.878 (0.042) [0.002]
Dep var mean	0.075	0.107	0.012	73.948	4.852	2.733
Bandwidth	395	300	381	274	289	283
Effective observations	29,139	21,411	28,229	42,896	45,059	44,130
	Avg. Educ	Avg. criminal record	Avg. assets (asinh)	General caste perc.	Female perc.	Avg. age
<i>Panel B. Candidate characteristics</i>						
RD_Estimate	0.017 (0.024) [0.611]	−0.018 (0.009) [0.044]	−0.014 (0.063) [0.754]	−0.016 (0.007) [0.028]	−0.021 (0.007) [0.004]	−0.095 (0.119) [0.611]
Dep var mean	2.312	0.137	12.604	0.264	0.431	41.117
Bandwidth	342	292	333	354	319	276
Effective observations	52,917	22,684	25,766	54,965	49,750	43,069
	Education	Criminal record	Total assets (asinh)	General caste	Female	Age
<i>Panel C. Winner characteristics</i>						
RD_Estimate	0.034 (0.034) [0.395]	−0.027 (0.011) [0.032]	−0.009 (0.084) [0.988]	−0.012 (0.007) [0.074]	−0.026 (0.008) [0.002]	−0.002 (0.186) [0.988]
Dep var mean	2.494	0.133	12.878	0.271	0.433	41.112
Bandwidth	290	229	327	368	328	298
Effective observations	66,416	17,913	25,322	85,818	76,287	69,302

Notes: This table reports regression discontinuity estimates from the main estimating equation of the effect of being above the treatment population threshold on electoral outcomes in that village's GP. The outcomes in panel A are related to the overall political participation in the GP, panel B relates to the characteristics of candidates contesting for the position of pradhan, and panel C focused on the characteristics of the candidate elected as pradhan. The data are from meetings notes of the Gram Panchayat Development Programme (GPDP) from the Ministry of the Panchayati Raj and the Uttar Pradesh State Election Commission for the 2015 and 2020 elections, including aggregate voting data and candidate affidavits. The first three outcomes in panel A are attendance rates at GPDP gram sabha meetings for the whole village, female residents, and scheduled caste residents. The second three are voter turnout, number of candidates per 1,000 residents of the GP, and effective candidates (receiving more than 5 percent of votes) per 1,000 residents in the 2015 and 2020 GP elections. The outcomes in panel B are the average education level, criminal record, age, and asset holdings (inverse hyperbolic sine) of candidates, as well as fraction of candidates who are general caste and female. The outcomes in panel C are the education level, criminal record, age, asset holdings (inverse hyperbolic sine), caste identity, and gender of the person elected pradhan. The running variable is the population of the village in the 2011 census. Each specification uses a linear polynomial, triangular kernel, and MSE-optimal bandwidth estimated following Calonico et al. (2017). Standard errors are clustered at the gram panchayat level and reported below the point estimates. *p*-values are reported within brackets.

goods provision (Fujiwara 2015; Miller, 2008). Online Appendix G supplements this by examining the correlation between these civic participation measures and the outcomes in our study, finding a robust positive relationship. Thus, it seems plausible that higher rates of civic participation explain some of our results, although we cannot quantify the magnitude.

B. Political Selection and Elite Capture

Another important political channel is selection and discipline of leaders. Seabright (1996) points out that leader actions are more observable in smaller communities, which can make it more costly to deviate from voter preferences. In smaller polities, it may also be easier to enact social sanctions on nonperforming leaders or observe candidate quality, which could generate better selection from within a given pool of candidates as well as incentivize better candidates (Dal Bó and Finan 2018). However, smaller polities tend to have fewer potential candidates, and so those who run may mechanically be lower average quality.

Panels B and C of Table 5 examine the characteristics of the individuals who run and win elections for pradhan positions. We use data from affidavits filed by candidates, as in other studies on Indian elections (e.g., Fisman, Schulz, and Vig 2014; Prakash, Rockmore, and Uppal 2019). For all three election cycles, the affidavits record the education, gender, age, and caste grouping (general caste, other backward class, scheduled caste, scheduled tribe) of candidates,³⁷ while for the 2020 elections, they also provide wealth and criminal records. Neither the average nor winning candidates' education is related to GP size (column 1). However, candidates and winners are 1.8 (14 percent; $p = 0.044$) and 2.7 (20.3 percent; $p = 0.032$) percentage points less likely to have a criminal record (column 2). Multiple recent studies have shown that Uttar Pradesh voters prefer candidates with no criminal record (Banerjee et al. 2014; George, Gupta, and Neggers 2019). These results are consistent with decentralization theories in which voters in smaller jurisdictions have an informational advantage in observing characteristics such as criminality (Boffa, Piolatto, and Ponzetto 2016) and deter such candidates from running. Prakash, Rockmore, and Uppal (2019) shows that criminally accused Indian politicians achieve worse economic outcomes—albeit for a higher level of office than studied here—so this could possibly explain some of the earlier results on service delivery.³⁸ However, our data do not permit direct tests of the informational channel or the effects of criminality on performance.

An additional aspect of political selection is the extent to which elites capture political offices and use them to pursue their personal interests. A major argument against decentralization has been that local governments could be more susceptible to capture by local elites (Bardhan and Mookherjee 2000, 2006). If clientelistic relationships of landowners determine capture in rural India (Anderson, Francois, and Kotwal 2015) and a smaller group own a greater fraction of land in smaller polities, capture may be easier. Measuring the extent of elite capture is typically difficult, but the affidavit data provide two excellent indicators: wealth and caste status. In contrast to those predictions, panel C of Table 5 finds that leaders elected in villages allocated to smaller local government units are no wealthier. Furthermore, candidates and elected leaders are actually less likely to be from higher castes ($p = 0.028$, $p = 0.074$).³⁹

³⁷ In the 2010 data, we only observe these for the winner of the election.

³⁸ Online Appendix G does find a negative correlation between politician criminality and some of the outcomes we study, but these relationships are not necessarily causal.

³⁹ This is in the context of reservation, where around half of elections can only be contested by candidates from low-caste backgrounds. Results on wealth are similar when we restrict to contests without reservation (panel C of

Another way of examining elite capture is to see who benefits from service delivery improvements. If there were substantial elite capture, benefits may primarily accrue to upper caste groups; indeed, B.R. Ambedkar, known as the father of the Indian Constitution, opposed the creation of decentralized village government due to concerns about capture by such groups. Instead, as discussed in Section IVC, online Appendix Table A9 shows that gains in workfare benefits are similar for both SC and non-SC. Section IVB also finds improvements in targeting, which is further inconsistent with more benefits accruing to elites. Those results are perhaps stronger evidence against elite capture than the findings on the identity of representatives, as they capture the consequences for citizens.⁴⁰ On the whole, we interpret these results as indicating a modest improvement in leader selection along the dimension of criminality, without compensating negative selection on other observable dimensions such as education or elite status. However, there are many dimensions of leader selection that we cannot observe, and so this may not provide the full picture.

C. Representative Incentives and Workload

Polity size can also affect leader incentives and workload. In a smaller polity, leader incentives may be better aligned with constituent preferences (Besley et al. 2004); in a larger polity, leaders have electoral incentives to follow the will of the majority of the population, which may lead them to put less effort into assisting smaller communities within the polity or to make decisions differing from the preferences of those communities. In the context of Uttar Pradesh, a village with a population to the right of the discontinuity at the time of delimitation will be split into its own polity and as a result, have a leader whose only electoral incentives are to serve that village. If the village were to the left of the discontinuity, it is agglomerated with other villages, and so the leader will have electoral incentives that are more divided. However, the consequences of these divided incentives depend on the relative populations of villages within the undivided GP. If the village of interest held the largest number of votes in the undivided GP, then leaders will have a stronger incentive to serve it than if it held a smaller fraction of the voters. If this mechanism is relevant, it would suggest that the treatment effect of a village being split into its own polity should be larger in cases where it would have been a political minority if it had not been split off.

To test this, we need to know the GP composition prior to delimitation, and so we focus on the 2015 delimitation. We split the sample based on whether the village was the largest village in their GP at the time of delimitation and reestimate the main RD specification for the 2015 delimitation exercise within each subsample.

online Appendix Table A12), and if anything, it looks like election winners may be less wealthy in unreserved seats. However, there is no longer any effect on caste, so the other result could be due to a higher fraction of small polities being reserved for low-caste candidates, as they are more likely to have a high fraction of lower-caste inhabitants (see Dunning and Nilekani 2013 for a description of caste reservations).

⁴⁰ Panel C, column 5 of Table 5 also shows that leaders in the smaller polities are less likely to be female, which could have negative consequences for service delivery, especially for female-focused services (Chattopadhyay and Duflo 2004; Beaman et al. 2009). However, these costs appear to be at least balanced out by the other positive mechanisms to produce net benefits. We do not see a differentially beneficial effect for males for the one outcome that is disaggregated by gender—person-days of work in NREGS (panel A, column 7 of online Appendix Table A9)—with an approximately 15 percent improvement for both men and women.

TABLE 6—HETEROGENEOUS EFFECTS BASED ON PAST GP COMPOSITION

	Largest village		Pradhan village		Caste homogeneity		Habitations	
	No	Yes	No	Yes	Decreases/Same	Increases	Single	Multiple
<i>Panel A. Service delivery performance</i>								
RD_Estimate	0.108 (0.040) [0.006]	0.061 (0.055) [0.263]	0.114 (0.048) [0.018]	0.060 (0.053) [0.262]	0.094 (0.043) [0.027]	0.077 (0.044) [0.083]	0.066 (0.040) [0.103]	0.093 (0.044) [0.035]
Equality test (<i>p</i> -value)		0.483		0.454		0.779		0.65
Dep var mean	0.080	0.151	0.069	0.181	0.102	0.109	0.092	0.119
Bandwidth	219	140	173	173	205	176	212	189
Effective observations	9,138	4,678	5,972	4,570	7,708	7,047	8,937	7,190
<i>Panel B. NREGS performance</i>								
RD_Estimate	0.287 (0.038) [0.000]	0.017 (0.054) [0.755]	0.213 (0.041) [0.000]	0.131 (0.048) [0.006]	0.142 (0.042) [0.001]	0.172 (0.036) [0.000]	0.144 (0.032) [0.000]	0.185 (0.037) [0.000]
Equality test (<i>p</i> -value)		0		0.197		0.596		0.403
Dep var mean	−0.012	0.275	−0.002	0.222	0.107	0.071	0.080	0.107
Bandwidth	294	205	289	324	274	355	437	349
Effective observations	54,800	29,805	45,224	36,151	45,729	62,451	83,702	57,088

Notes: This table reports regression discontinuity estimates from the main estimating equation of the effect of being above the treatment population threshold on different welfare program and NREGS outcomes. The outcome in panel A is an index of number of beneficiaries for welfare programs as measured in the Mission Antyodaya data in 2019, while panel B is an index of NREGS program outcomes, both as calculated following Kling, Katz, and Liebman (2001). Columns 1 and 2 split the sample based on whether a village was the largest village in its GP prior to 2015 and report the RD estimates from each subsample. Columns 3 and 4 split the sample based on whether the village was the home village of the pradhan during the 2010–2015 period, prior to the 2015 delimitation. Columns 5 and 6 split the sample based on whether splitting a village into its own GP would increase or decrease caste fractionalization in its GP. Columns 7 and 8 split the sample based on whether the village has a single habitation or multiple habitations. See the online Appendix for details of each variable. The running variable is the population of the village in the 2011 census. Each specification uses a linear polynomial, triangular kernel, and MSE-optimal bandwidth estimated following Calonico et al. (2017). Standard errors are clustered at the gram panchayat level and reported below the point estimates. *p*-values are reported within brackets.

Columns 1 and 2 of Table 6 examine this heterogeneity with the indexes of outcomes from Table 3 (individually targeted programs; workfare). In panel A, the coefficient is larger and only statistically significant for villages that were not previously the largest village in their GP, but the difference in coefficients is not statistically significant ($p = 0.48$). However, in the case of workfare performance, we strongly reject equivalence of the coefficients ($p < 0.001$). Effects are much stronger for villages that were not previously the largest village in their GP, where the point estimate is close to zero for villages that were previously the largest village.⁴¹

⁴¹ These results also help in understanding whether the increases in outcomes in Section IV reflect a zero-sum competition between villages within a single GP. To fix ideas, suppose that there are two villages in a GP, labeled *a* and *b* with populations $p_a + p_b = P$. Suppose also that resource allocations were proportional to population across GPs, such that a GP containing both villages receives tP , where t is the transfer per person. If $p_a > p_b$, and so village *a* dominates politically, then village *a* may receive more than $\frac{p_a}{p_a + p_b}tP$. This means that village *b* would benefit from being split into its own GP because it would get $t p_b$, but village *a* would lose resources and thus have worse outcomes. The data are not consistent with this explanation; instead, service delivery outcomes are no worse in villages that were previously the largest village

An alternative interpretation of these results is that the *pradhan* is more likely to come from the most populous village within a GP, as that village will have the votes to dominate elections; indeed, the *pradhan* is elected from the largest village in 80.2 percent of multivillage GPs. The observed heterogeneity could thus be about having locally based leadership independent of electoral incentives. In particular, such a local leader may have better information about the local population for targeting of state interventions (Dal Bó et al. 2021; Balán et al. 2022), greater vulnerability to informal sanctions for poor performance, or stronger prosocial preferences toward their own community. We provide a test of this interpretation using data on the home address of the *pradhan* from the 2010 GP elections. We follow the same approach as with the largest village analysis but now split the sample according to whether the 2010–2014 *pradhan* came from the village. Columns 3 and 4 of Table 6 do not find statistically significant heterogeneity related to the *pradhan*'s village of residence.⁴² These findings thus suggest that having a locally based leader is not the driving force behind the largest village effects, and so our estimates are more consistent with electoral incentives.⁴³

Another aspect of representative functionality is their workload. The observed benefits of smaller polities could reflect a higher ratio of representatives to citizens; when serving a smaller group, it is the same cost for a representative to provide a higher per person level of service. In the context of the gram panchayat, there are three representatives who may be relevant: the panchayat secretary, who is the bureaucratic counterpart to the *pradhan* and carries out village-level bureaucratic tasks; ward councilors, who each represent a jurisdiction of a few hundred people on the village council; and the village head/*pradhan*. In online Appendix G, we use personnel records to show that villages to the right of the discontinuity only have a 4 percent better panchayat secretary:citizen ratio; this is because a single secretary typically serves many GPs (online Appendix Table A14). It further estimates that this reduction in secretary workload is unlikely to explain more than 2.4 percent of the overall treatment effect, so we do not think this is a major driver of our effects. We also use a separate regression discontinuity design to show that a higher ratio of ward councilors to citizens has no effect on our outcomes, and so reduced ward councilor workload cannot explain the results.

We do not have a good way of assessing the importance of lower *pradhan* workload. However, most forms of local government globally have a unitary executive head. Reducing the workload of that head is therefore a highly generalizable part of the creation of smaller polities even if we cannot quantify its importance. Thus, if this mechanism is important, our results still should generalize to other contexts.⁴⁴

⁴²The difference in coefficients is reasonably large in panel B, but this is because of the correlation of *pradhan* village and largest village. When we hold largest village status fixed (i.e., compare effects across *pradhan*/non-*pradhan* villages among villages that were the largest in their GP), that difference disappears.

⁴³Online Appendix Table A13 examines how polity size affects the electoral competition. Elections are less competitive in smaller polities, where the average number of candidates declines (columns 1–2), a Herfindahl index of vote concentration increases (column 3), and there are fewer close elections (columns 4–7). It may be that it is easier for voters to coordinate on candidates when they have more information on them or preferences are more homogeneous in smaller polities, but our data do not permit definitive tests.

⁴⁴However, a piece of evidence against this as an important mechanism is that it would predict improvements even in villages that were previously the largest village in their GP, which we do not see (panel B, column 2 of Table 6).

This section offers evidence that the incentives of representatives play a role in the effect of polity size. An advantage of these tests is that they directly link the mechanisms to the outcomes on which we observed effects in Section IV. However, a village's political weight within its GP could be correlated with other, unobservable dimensions of heterogeneity, so this is more suggestive than definitive evidence.

D. Homogeneity and Geographical Compactness

Given that individuals tend to live near others from the same identity group, the creation of smaller local government units will tend to imply less diversity in the identity groups of residents. There may be more cooperation in more homogeneous jurisdictions (Anderson 2011; Bazzi and Gudgeon 2021), or leaders may be more likely to be of the same identity group as members of the population and thus more likely to help them (Neggers 2018; Sharan and Kumar 2021). We exploit variation in the extent to which the caste composition of a village's local government is affected by delimitation. In some cases, a pre-delimitation GP contained villages with different caste compositions, and so splitting off a village creates a more homogeneous polity; in other cases, the component villages of the pre-delimitation GP were similar, and so caste homogeneity does not change with a split. For this analysis, we can only use the post-2015 data since it requires knowing the pre-delimitation caste composition of the GP. In 2011, we observe the population in each village who are scheduled caste, scheduled tribe, or from neither group. We calculate fractionalization (Alesina et al. 2003; Bazzi et al. 2019) by caste in the village as well as in the pre-2015 GP. We then determine whether fractionalization would increase if the village were made into its own local government and test for heterogeneity along this margin.⁴⁵

In columns 5 and 6 of Table 6, we split villages based on whether delimitation would result in more (column 5) or less fractionalization (column 6) and rerun the main RD design in each subsample. We cannot reject equivalence of the estimates across these subsamples ($p = 0.779$ for service delivery in panel A; $p = 0.596$ for workfare in panel B), indicating that this dimension of heterogeneity is not a strong mediator of the observed effects, albeit with the caveat that this is not the only possible way to measure homogeneity.⁴⁶

Another consequence of creating smaller local governments is that they cover smaller geographical areas. Greater compactness may increase observability of leader actions and thus promote greater accountability. As with identity group homogeneity, we exploit heterogeneity in the extent to which the delimitation rule increases compactness. A village is often composed of multiple geographically

⁴⁵ It would be ideal to observe more precise caste categories (*jati*) for this calculation. However, even this aggregation should pick up one of the key political divisions in Uttar Pradesh, where one of the most important political parties in this time period had its base among scheduled castes (Bahujan Samaj Party).

⁴⁶ One commonly cited reason for creation of more homogeneous polities is that these will tend to have more homogeneous preferences even outside of identity group (Oates 1972). In democratic contexts, such polities may be easier to govern, as it is easier to aggregate voter preferences. In online Appendix G, we use the Rural Economic and Demographic Survey (REDS) to demonstrate that preferences indeed tend to be more homogeneous in smaller polities. However, we lack village-level survey data on preferences for our sample that would allow us to conduct a similar exercise to that with caste-based fractionalization, that is, check whether the treatment effects are greater in cases where delimitation increases preference homogeneity in the resulting polities.

distinct clusters of households known as “hamlets,” and so even a single village GP may still have distinct and geographically dispersed hamlets. If geographical compactness is an important mechanism, then the estimated effects should be largest when a village contains a single hamlet, as being split off will create more geographical compactness. Columns 7 and 8 of Table 6 split the sample by whether the village is a single hamlet or multi-hamlet village. We cannot reject the equality of the coefficients ($p = 0.65$ for service delivery; $p = 0.403$ for workfare), providing evidence against geographical compactness of the GP as a mechanism.⁴⁷ Note also that identity group homogeneity and geographical compactness are treatments that are necessarily bundled with reductions in polity population. The tests of this section thus not only speak to those two mechanisms but also suggest that population or other aspects of the treatment bundle drive our results rather than homogeneity or compactness.

VI. Conclusion

Ever since Plato, thinkers have debated the optimal size of political jurisdictions. In recent years, popular sentiment has moved in favor of smaller polities, but the relationship between polity size and governance outcomes is theoretically unclear. We find that smaller polities indeed provide more public services, but with meaningful heterogeneity in effects related to jurisdiction size. We also find evidence that civic engagement, leader selection, and political incentives all may play some role in the observed increases in services. While these findings are most directly applicable to the design of the local governments serving hundreds of millions of people in north India, these political mechanisms plausibly generalize to other democratic states with similar local government structures.

Our analysis has focused on increases in publicly provided benefits, but these increases have costs. In online Appendix F, we conduct a cost-benefit analysis and estimate that creating a polity has a direct cost of around \$2 per resident per year due to infrastructure and staffing costs. This is less than the sum of benefits accruing to residents from better service access, but policymakers must also consider the costs of providing those services. In the case of rural Uttar Pradesh, where public goods are notoriously underprovided, these investments likely have returns that exceed their costs in terms of future taxation. This will not necessarily be true in settings where public goods provision is already high, as a proliferation of smaller polities could lead to overprovision. Yet in either case, our results are still informative in measuring the consequences of polity size reforms, and it is a political decision whether the costs are worth the benefits.

Our data do not permit an evaluation of every possible mechanism through which polity size can affect public service delivery. Future research should consider alternate channels that we could not fully test here, such as workload of representatives, yardstick competition, heterogeneity of political preferences, and corruption, as well as how polity size interacts with other elements of the environment. It may also be

⁴⁷ Since this is a property of the village rather than the GP, we can use this approach with both the 1995 and 2015 delimitation episodes. Online Appendix Table A16 reruns this analysis with the post-1995 delimitation outcomes and similarly does not find evidence of geographical compactness mediating the effects.

that mechanisms that turn out not to be relevant in our context, such as elite capture, have relevance in environments with different institutional features. Understanding these nuances can help in better design of local government systems and is an exciting area for future research.

REFERENCES

- Alatas, Vivi, Abhijit Banerjee, Rema Hanna, Benjamin A. Olken, and Julia Tobias. 2012. "Targeting the Poor: Evidence from a Field Experiment in Indonesia." *American Economic Review* 102 (4): 1206–40.
- Alesina, Alberto, Reza Baqir, and William Easterly. 1999. "Public Goods and Ethnic Divisions." *Quarterly Journal of Economics* 114 (4): 1243–84.
- Alesina, Alberto, Arnaud Devleeschauwer, William Easterly, Sergio Kurlat, and Romain Wacziarg. 2003. "Fractionalization." *Journal of Economic Growth* 8 (2): 155–94.
- Alesina, Alberto, and Enrico Spolaore. 1997. "On the Number and Size of Nations." *Quarterly Journal of Economics* 112 (4): 1027–56.
- Alesina, Alberto, and Enrico Spolaore. 2005. *The Size of Nations*. Cambridge, MA: MIT Press.
- Alsop, R., Anirudh Krishna, and Disa Sjoblom. 2000. *Overview of Rural Decentralization in India*. Washington, DC: World Bank.
- Ambedkar, B. R. 1932. "Speech on Village Panchayats Bill in Bombay Legislature."
- Anderson, Siwan. 2011. "Caste as an Impediment to Trade." *American Economic Journal: Applied Economics* 3 (1): 239–63.
- Anderson, Siwan, Patrick Francois, and Ashok Kotwal. 2015. "Clientelism in Indian villages." *American Economic Review* 105 (6): 1780–1816.
- Asher, Sam, Tobias Lunt, Ryu Matsuura, and Paul Novosad. 2021. "Development Research at High Geographic Resolution." Unpublished.
- Asher, Sam, and Paul Novosad. 2020. "Rural Roads and Local Economic Development." *American Economic Review* 110 (3): 797–823.
- Balán, Pablo, Augustin Bergeron, Gabriel Tourek, and Jonathan L. Weigel. 2022. "Local Elites as State Capacity: How City Chiefs Use Local Information to Increase Tax Compliance in the Democratic Republic of the Congo." *American Economic Review* 112 (3): 762–97.
- Ban, Radu, Saumitra Jha, and Vijayendra Rao. 2012. "Who has Voice in a Deliberative Democracy? Evidence from Transcripts of Village Parliaments in South India." *Journal of Development Economics* 99 (2): 428–38.
- Banerjee, Abhijit, Donald P. Green, Jeffery McManus, and Rohini Pande. 2014. "Are Poor Voters Indifferent to Whether Elected Leaders are Criminal or Corrupt? A Vignette Experiment in Rural India." *Political Communication* 31 (3): 391–407.
- Banerjee, Abhijit, Esther Dufo, Clement Imbert, Santhosh Mathew, and Rohini Pande. 2020. "E-governance, Accountability, and Leakage in Public Programs: Experimental Evidence from a Financial Management Reform in India." *American Economic Journal: Applied Economics* 12 (4): 39–72.
- Bardhan, Pranab. 2002. "Decentralization of Governance and Development." *Journal of Economic Perspectives* 16 (4): 185–205.
- Bardhan, Pranab, and Dilip Mookherjee. 2000. "Capture and Governance at Local and National Levels." *American Economic Review* 90 (2): 135–39.
- Bardhan, Pranab, and Dilip Mookherjee. 2006. "Decentralisation and Accountability in Infrastructure Delivery in Developing Countries." *Economic Journal* 116 (508): 101–27.
- Bazzi, Samuel, Arya Gaduh, Alexander D. Rothenberg, and Maisy Wong. 2019. "Unity in Diversity? How Intergroup Contact Can Foster Nation Building." *American Economic Review* 109 (11): 3978–4025.
- Bazzi, Samuel, and Matthew Gudgeon. 2021. "The Political Boundaries of Ethnic Divisions." *American Economic Journal: Applied Economics* 13 (1): 235–66.
- Beaman, Lori, Raghavendra Chattopadhyay, Esther Dufo, Rohini Pande, and Petia Topalova. 2009. "Powerful Women: Does Exposure Reduce Bias?" *Quarterly Journal of Economics* 124 (4): 1497–1540.
- Besley, Timothy, and Anne Case. 1995. "Incumbent Behavior: Vote-Seeking, Tax-Setting, and Yardstick Competition." *American Economic Review* 85 (1): 25–45.
- Besley, Timothy, Rohini Pande, Lupin Rahman, and Vijayendra Rao. 2004. "The Politics of Public Good Provision: Evidence from Indian Local Governments." *Journal of the European Economic Association* 2 (2-3): 416–26.

- Boffa, Federico, Amedeo Piolatto, and Giacomo A. M. Ponzetto.** 2016. "Political Centralization and Government Accountability." *Quarterly Journal of Economics* 131 (1): 381–422.
- Burlig, Fiona, and Louis Preonas.** 2021. "Out of the Darkness and into the Light? Development Effects of Rural Electrification." Unpublished.
- Bussell, Jennifer.** 2019. *Clients and Constituents: Political Responsiveness in Patronage Democracies*. Oxford: Oxford University Press.
- Calonico, Sebastian, Matias D. Cattaneo, and Max H. Farrell.** 2018. "On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference." *Journal of the American Statistical Association* 113 (522): 767–79.
- Calonico, Sebastian, Matias D. Cattaneo, and Rocio Titiunik.** 2014. "Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs." *Econometrica* 82 (6): 2295–2326.
- Calonico, Sebastian, Matias D. Cattaneo, Max H. Farrell, and Rocio Titiunik.** 2017. "rdrobust: Software for Regression-Discontinuity Designs." *Stata Journal* 17 (2): 372–404.
- Cassidy, Travis, and Tejaswi Velayudhan.** 2022. "Government Fragmentation and Economic Growth." Unpublished.
- Cattaneo, Matias D., Michael Jansson, and Xinwei Ma.** 2018. "Manipulation Testing Based on Density Discontinuity." *Stata Journal* 18 (1): 234–61.
- Cattaneo, Matias D., Michael Jansson, and Xinwei Ma.** 2020. "Simple Local Polynomial Density Estimators." *Journal of the American Statistical Association* 115 (531): 1449–55.
- Centre for Policy Research.** 2019. *Analysis of Fund Flows to Rural Local Bodies—A study for the Fifteenth Finance Commission*. New Delhi: Centre for Policy Research.
- Chakraborty, Pinaki, Manish Gupta, and Rajesh Thakur.** 2018. *Overview of State Finance Commission Reports*. New Delhi: National Institute of Public Finance and Policy.
- Chattopadhyay, Raghavendra, and Esther Duflo.** 2004. "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India." *Econometrica* 72 (5): 1409–43.
- Chauchard, Simon.** 2017. *Why Representation Matters: The Meaning of Ethnic Quotas in Rural India*. Cambridge: Cambridge University Press.
- Chaudhary, Latika, and Lakshmi Iyer.** 2022. "It Takes a Village? Administrative Devolution and Human Development in India." Unpublished.
- Coady, David, Margaret E. Grosh, and John Hoddinott.** 2004. *Targeting of Transfers in Developing Countries: Review of Lessons and Experience*. Washington, DC: World Bank.
- Cohen, Isabelle.** Forthcoming. "Documenting Decentralization: Empirical Evidence on Administrative Unit Proliferation from Uganda." *World Bank Economic Review*.
- Cremer, Jacques, Antonio Estache, and Paul Seabright.** 1994. "The Decentralization of Public Services: Lessons from the Theory of the Firm." Unpublished.
- Crépon, Bruno, Esther Duflo, Marc Gurgand, Roland Rathelot, and Philippe Zamora.** 2013. "Do Labor Market Policies have Displacement Effects?" *Quarterly Journal of Economics* 128 (2): 531–80.
- Dahis, Ricardo, and Christiane Szerman.** 2021. "Development via Administrative Redistricting: Evidence from Brazil." Unpublished.
- Dal Bó, Ernesto, and Frederico Finan.** 2018. "Progress and Perspectives in the Study of Political Selection." *Annual Review of Economics* 10: 541–75.
- Dal Bó, Ernesto, Frederico Finan, Nicholas Y. Li, and Laura Schechter.** 2021. "Information Technology and Government Decentralization: Experimental Evidence from Paraguay." *Econometrica* 89 (2): 677–701.
- Dal Bó, Ernesto, and Martín A. Rossi.** 2011. "Term Length and the Effort of Politicians." *Review of Economic Studies* 78 (4): 1237–63.
- Das, Sabyasachi.** 2015. "Efficacy of 'Town Hall' Meetings in Electoral Democracy: Theory and Evidence from Indian Village Councils." Unpublished.
- Department of Drinking Water and Sanitation.** 2011. *Jal Jeevan Village Directory*. Ministry of Jal Shakti.
- Dunning, Thad, and Janhavi Nilekani.** 2013. "Ethnic Quotas and Political Mobilization: Caste, Parties, and Distribution in Indian Village Councils." *American Political Science Review* 107 (1): 35–56.
- Faguet, Jean-Paul.** 2004. "Does Decentralization Increase Government Responsiveness to Local Needs? Evidence from Bolivia." *Journal of Public Economics* 88 (3–4): 867–93.
- Fisman, Raymond, Florian Schulz, and Vikrant Vig.** 2014. "The Private Returns to Public Office." *Journal of Political Economy* 122 (4): 806–62.
- Fujiwara, Thomas.** 2015. "Voting Technology, Political Responsiveness, and Infant Health: Evidence from Brazil." *Econometrica* 83 (2): 423–64.
- Gautam, Sanghmitra.** 2023. "Quantifying Welfare Effects in the Presence of Externalities: An Ex-ante Evaluation of Sanitation Interventions." *Journal of Development Economics* 164: 103083.

- Gelman, Andrew, and John Carlin. 2014. "Beyond Power Calculations: Assessing Type S (Sign) and Type M (Magnitude) Errors." *Perspectives on Psychological Science* 9 (6): 641–51.
- Genicot, Garance, Cait Brown, and Nishtha Kochhar. 2021. "Political Reservations as Term-Limits." Unpublished.
- George, Siddharth, Sarika Gupta, and Yusuf Neggers. 2019. "Can We Text Criminal Politicians Out of Office? Evidence from a Mobile Experiment in India." Unpublished.
- Gerring, John, Maxwell Palmer, Jan Teorell, and Dominic Zarecki. 2015. "Demography and Democracy." *American Political Science Review* 109 (3): 574–91.
- Geruso, Michael, and Dean Spears. 2018. "Neighborhood Sanitation and Infant Mortality." *American Economic Journal: Applied Economics* 10 (2): 125–62.
- Gómez-Reino, Juan Luis, Santiago Lago-Peñas, and Jorge Martínez-Vazquez. 2021. "Evidence on Economies of Scale in Local Public Service Provision: A Meta-Analysis." *Journal of Regional Science* 63 (4): 793–819.
- Government of India. 1991, 2001, 2011. *Census of India*. Office of the Registrar General and Census Commissioner. <https://censusindia.gov.in/census.website/data/census-tables>.
- Gulati, Ashok, and Shweta Saini. 2015. "The Public Distribution System in India: Policy Evolution and Program Delivery Trends." *Food Policy* 55: 77–89.
- Holmström, Bengt, and Paul Milgrom. 1991. "Multitask Principal–Agent Analyses: Incentive Contracts, Asset Ownership, and Job Design." *Journal of Law, Economics, and Organization* 7: 24–52.
- Indian Institute of Public Administration. 2020. *Measures to Augment the Resources of Panchayats: Empirical Assessment 2018-19 (Annex IX: A Report on Uttar Pradesh)*. New Delhi: Indian Institute of Public Administration.
- Jha, Rajesh. 2021. "Social-Objectives-Laden Electoral Rules of Panchayati Raj Institutions: A Political Justice Perspective." *JournalNX* 7 (6): 425–33.
- Khanna, Gaurav. 2023. "Large-Scale Education Reform in general Equilibrium: Regression Discontinuity Evidence from India." *Journal of Political Economy* 131 (2): 549–91.
- Kingdon, Geeta, and Mohd Muzammil. 2009. "A Political Economy of Education in India: The Case of Uttar Pradesh." *Oxford Development Studies* 37 (2): 123–44.
- Kling, Jeffrey R., Lawrence F. Katz, and Jeffrey B. Liebman. 2001. "Moving to Opportunity in Boston: Early Results of a Randomized Mobility Experiment." *Quarterly Journal of Economics* 116 (2): 607–54.
- Kling, Jeffrey R., Jeffrey B. Liebman, and Lawrence F. Katz. 2007. "Experimental Analysis of Neighborhood Effects." *Econometrica* 75 (1): 83–119.
- Kruks-Wisner, Gabrielle. 2018. *Claiming the State: Active Citizenship and Social Welfare in Rural India*. Cambridge: Cambridge University Press.
- Lassen, David Dreyer, and Søren Serritzlew. 2011. "Jurisdiction Size and Local Democracy: Evidence on Internal Political Efficacy from Large-Scale Municipal Reform." *American Political Science Review* 105 (2): 238–58.
- Lewis-Faupel, Sean, Yusuf Neggers, Benjamin A. Olken, and Rohini Pande. 2016. "Can Electronic Procurement Improve Infrastructure Provision? Evidence from Public Works in India and Indonesia." *American Economic Journal: Economic Policy* 8 (3): 258–83.
- Lipscomb, Molly, and Ahmed Mushfiq Mobarak. 2016. "Decentralization and Pollution Spillovers: Evidence from the Re-Drawing of County Borders in Brazil." *Review of Economic Studies* 84 (1): 464–502.
- Mansuri, Ghazala, and Vijayendra Rao. 2013. *Localizing Development: Does Participation Work?* Washington, DC: World Bank.
- Martínez-Bravo, Monica. 2014. "The Role of Local Officials in New Democracies: Evidence from Indonesia." *American Economic Review* 104 (4): 1244–87.
- Martínez-Bravo, Monica. 2017. "The Local Political Economy Effects of School Construction in Indonesia." *American Economic Journal: Applied Economics* 9 (2): 256–89.
- Martínez-Bravo, Monica, Gerard Padró i Miquel, Nancy Qian, and Yang Yao. 2022. "The Rise and Fall of Local Elections in China." *American Economic Review* 112 (9): 2921–58.
- Miller, Grant. 2008. "Women's Suffrage, Political Responsiveness, and Child Survival in American History." *Quarterly Journal of Economics* 123 (3): 1287–1327.
- Ministry of the Panchayati Raj. 2016. *Devolution Report 2015–2016: Where Local Democracy and Devolution in India is Heading Towards*. New Delhi: Ministry of the Panchayati Raj.
- Ministry of the Panchayat Raj. 2012–2020. Gram Panchayat Expenditure Data. <https://egramswaraj.gov.in/>.
- Ministry of the Panchayat Raj. 2020. *Local Government Directory*. <https://lgdirectory.gov.in/>.
- Ministry of the Panchayat Raj and Ministry of Rural Development. 2022–2023. Gram Sabha Meeting Data, Panchayat Development Plan. <https://gdpd.nic.in/>.

- Ministry of Rural Development.** 2011–2012. *Socio-Economic and Caste Census*. <https://rural.gov.in/en/secc-2011>.
- Ministry of Rural Development.** 2016–2020. *NREGS Program Data*. https://mnregaweb4.nic.in/netnrega/dynamic2/dynamicreport_new4.aspx.
- Ministry of Rural Development.** 2019. *Mission Antyodaya 2019*. <https://missionantyodaya.nic.in>.
- Mookherjee, Dilip.** 2015. “Political Decentralization.” *Annual Review of Economics* 7 (1): 231–49.
- Narasimhan, Veda.** 2021. “Decentralization, Rural Development, and Growth: Evidence from India.” Unpublished.
- Narasimhan, Veda, and Jeffrey Weaver.** 2023. “Political Representation and Public Goods Provision.” Unpublished.
- Narasimhan, Veda, and Jeffrey Weaver.** 2024. *Data and Code for: “Polity Size and Local Government Performance: Evidence from India.”* Nashville, TN: American Economic Association; distributed by Inter-university Consortium for Political and Social Research, Ann Arbor, MI. <https://doi.org/10.3886/E202444V1>.
- Neggers, Yusuf.** 2018. “Enfranchising Your Own? Experimental Evidence on Bureaucrat Diversity and Election Bias in India.” *American Economic Review* 108 (6): 1288–1321.
- NITI Aayog.** 2021. *Multi-dimensional Poverty Index. NITI report*.
- Oates, Wallace.** 1972. *Fiscal Federalism*. New York, NY: Harcourt.
- Olken, Benjamin A.** 2010. “Direct Democracy and Local Public Goods: Evidence from a Field Experiment in Indonesia.” *American Political Science Review* 104 (2): 243–67.
- Pierskalla, Jan H.** 2016. “Splitting the Difference? The Politics of District Creation in Indonesia.” *Comparative Politics* 48 (2): 249–68.
- Planning Commission of India.** 2013. *Evaluation Study on Total Sanitation Campaign*. New Delhi: Planning Commission of India.
- Prakash, Nishith, Marc Rockmore, and Yogesh Uppal.** 2019. “Do Criminally Accused Politicians Affect Economic Outcomes? Evidence from India.” *Journal of Development Economics* 141: 102370.
- Rao, Govinda.** 2007. “Rural fiscal decentralization in India: Problems and reform issues.” *Decentralization: Institutions and Politics in Rural India* 135–53.
- Rao, Vijayendra, and Paromita Sanyal.** 2010. “Dignity through Discourse: Poverty and the Culture of Deliberation in Indian Village Democracies.” *Annals of the American Academy of Political and Social Science* 629 (1): 146–72.
- Seabright, Paul.** 1996. “Accountability and Decentralisation in Government: An Incomplete Contracts Model.” *European Economic Review* 40 (1): 61–89.
- Sethi, Geeta, et al.** 2004. *Fiscal Decentralization to Rural Governments in India*. Oxford: Oxford University Press.
- Sharan, M. R., and Chinmaya Kumar.** 2021. “Something to Complain About: How Minority Representatives Overcome Ethnic Differences.” Unpublished.
- Singhania, Deepak.** 2022. “Do Smaller Local Governments Bring Citizens More? Evidence from Direct Elections in Indonesia.” Unpublished.
- Spears, Dean, and Sneha Lamba.** 2016. “Effects of Early-Life Exposure to Sanitation on Childhood Cognitive Skills.” *Journal of Human Resources* 51 (2): 298–327.
- Tiebout, Charles M.** 1956. “A Pure Theory of Local Expenditures.” *Journal of Political Economy* 64 (5): 416–24.
- Uttar Pradesh State Election Committee.** 2010, 2015, 2020. *Panchayat Election Data*. <https://sec.up.nic.in>.
- Vaishnav, Milan.** 2017. *When Crime Pays: Money and Muscle in Indian Politics*. New Haven, CT: Yale University Press.
- Vaishnav, Milan, and Neelanjan Sircar.** 2012. “Core or Swing? The Role of Electoral Context in Shaping Pork Barrel.” Unpublished.
- Weaver, Jeffrey.** 2021. “Political Decentralization and Public Goods Provision.” Unpublished.
- Wilkinson, Steven I.** 2006. “The Politics of Infrastructural Spending in India.” Unpublished.